

Selecting Blockchain Networks for Tokenized Real-World Assets:

A Systematic Review and Requirements-Based Decision Framework

Rischan Mafrur ¹ and Khadijah²

¹ Western Sydney University, Surabaya Campus, R.Mafrur@westernsydney.edu.au

² CryptocoinHalal.com, khadijah@cryptocoinhalal.com

July 24, 2026

Abstract

Tokenized real-world assets (RWAs) are often discussed as if blockchain selection were primarily a question of transaction speed, fees, or ecosystem size. That framing is incomplete. For tokenized Treasuries, bonds, private credit, funds, equities, commodities, and real estate, the decisive issue is not whether a blockchain is *technically usable*, but whether the complete issuance stack can support the legal claim, compliance model, custody structure, transfer constraints, settlement design, redemption process, market access, and proof-of-backing required by the instrument. This article conducts a systematic review informed by PRISMA 2020 reporting principles. Approximately 320 potentially relevant records were identified across academic, regulatory, institutional, technical, and official market-infrastructure sources published between 2015 and July 2026. Following deduplication, relevance screening, and source-quality appraisal, 158 unique sources were retained for the final synthesis. Using Task–Technology Fit as the principal theoretical lens, the review conceptualizes tokenization as a set of tasks (i.e., issuing, managing, transferring, settling, redeeming, and governing RWAs) and evaluates fit against the capabilities of the full issuance stack. The review synthesizes five requirement dimensions and develops a two-stage decision framework. Stage 1 applies mandatory feasibility gates to exclude issuance stacks that cannot satisfy minimum legal, regulatory, operational, privacy, settlement, redemption, or proof-of-backing conditions. Stage 2 compares only feasible alternatives using qualitative criteria such as security, finality, cost, interoperability, liquidity infrastructure, stablecoin support, developer support, and operational complexity. The central conclusion is that no blockchain network is universally best for RWA tokenization; suitability is contingent on the asset, jurisdiction, structure, and supporting institutional infrastructure.

Keywords: real-world asset tokenization; blockchain selection; digital securities; tokenized funds; tokenized Treasuries; custody; settlement; compliance; Task–Technology Fit; systematic review

1 Introduction

Tokenization has moved from conceptual experimentation toward production use in selected segments of capital markets, especially money market funds, government debt exposure, selected bond pilots, and tightly controlled private-market products (Bank for International Settlements 2023; Aldasoro et al. 2023; Committee on Payments and Market Infrastructures 2024; International Organization of Securities Commissions 2025; Financial Stability Board 2024; Organisation for Economic Co-operation and Development 2025; Adrian 2026). Yet the debate about “which blockchain to use” often remains technologically reductionist. Market commentary frequently elevates throughput, transaction fees, total value locked (TVL), or chain popularity as if they were sufficient decision variables. That is analytically weak for RWAs, because tokenization normally represents legal or contractual claims that remain embedded in off-chain institutions, registries, custodial arrangements, transfer rules, payment rails, and supervisory expectations (Konashevych 2020; World Bank 2020; Kaousar Nassr 2021; International Organization of Securities Commissions 2025; U.S. Securities and Exchange Commission 2026).

The problem is especially acute when comparing heterogeneous products under a single label. A tokenized Treasury exposure distributed only to qualified purchasers is not equivalent to a retail-accessible commodity token, a permissioned private-credit instrument, or a tokenized public-equity share class. The legal authority of the ownership record, the need for transfer-agent participation, the requirements for investor eligibility, the existence of clawback or forced-transfer powers, and the operational importance of settlement and redemption differ materially across instruments (Monetary Authority of Singapore 2025b; Monetary Authority of Singapore 2025a; U.S. Securities and Exchange Commission 2025c; Investor.gov 2025; European Union 2022; European Union 2023). Consequently, a blockchain network cannot be evaluated in isolation. The relevant analytical unit is the *issuance stack*: blockchain network, token standard, identity and compliance layer, smart contracts, custodian, transfer agent or fund administrator, settlement and payment infrastructure, oracle and proof-of-backing layer, and trading or liquidity infrastructure.

This article addresses that gap through a systematic integrative review and develops a requirements-based decision framework for selecting suitable blockchain networks and issuance infrastructures for tokenized RWAs. The framework does *not* ask which chain is universally best. Instead, it asks which issuance-stack configurations are feasible and preferable for a specified RWA under stated legal, regulatory, operational, market, and technical constraints. The review responds to six research questions:

- RQ1.** What legal, regulatory, operational, technical, market, and evidentiary requirements influence the selection of issuance stacks for tokenized RWAs?
- RQ2.** How do these requirements differ across RWA classes and tokenization structures?
- RQ3.** At which layers of the issuance stack are the required capabilities delivered?
- RQ4.** Which requirements should operate as mandatory feasibility gates?
- RQ5.** How should feasible issuance stacks be compared after all mandatory requirements have been satisfied?

RQ6. How does the framework affect architecture selection when applied to an institutional private-credit use case?

The article makes four contributions. First, it consolidates fragmented literature across blockchain, fintech, information systems, securities regulation, market infrastructure, and practitioner documentation into a unified requirements taxonomy. Second, it distinguishes native blockchain capabilities from those delivered through token standards, smart contracts, middleware, or external institutions. Third, it applies Task–Technology Fit (TTF) as the principal explanatory theory, supplemented where useful by contingency and institutional lenses, to explain why fit matters more than generic network performance (Goodhue and Thompson 1995; Zigurs and Buckland 1998; Dishaw and Strong 1999; Donaldson 2001; DiMaggio and Powell 1983; Tornatzky and Fleischer 1990). Fourth, it offers a two-stage decision framework, an issuer decision tree, a due-diligence checklist, and a worked application involving an institutional private-credit product.

2 RWA tokenization background

RWA tokenization is best understood as the creation of digital tokens that either constitute, evidence, mirror, or help operationalize rights to assets or cash flows grounded in the off-chain economy. In finance, those assets may include government debt, corporate bonds, fund interests, equity securities, commodities, warehouseed goods, real estate interests, receivables, and private-credit claims (Kaousar Nassr 2021; Committee on Payments and Market Infrastructures 2024; International Organization of Securities Commissions 2025; Luo et al. 2026; Vella et al. 2026). In law and operations, however, “tokenization” does not by itself answer what the token legally is. A token may represent a direct issued instrument on a ledger, a digitally mirrored share class, a contractual claim on a special-purpose vehicle, a fund share, a warehouse-receipt-like claim, or an entitlement administered by a platform operator (U.S. Securities and Exchange Commission 2026; Konashevych 2020; World Bank 2020; Organisation for Economic Co-operation and Development 2021).

That distinction has practical significance because “on-chain visibility” is not equivalent to legal certainty. A blockchain can expose token balances, transfer histories, and smart-contract rules, but it cannot independently prove the existence, ownership, quality, absence of encumbrance, valuation, or redemption availability of off-chain assets. Those matters depend on legal documentation, custody arrangements, fund governance, trustee or transfer-agent processes, administrator calculations, auditor or attestor evidence, and reliable data flows from off-chain institutions (Caldarelli 2020; Al-Breiki et al. 2020; Chainlink 2025b; Chainlink 2025a; Paxos 2025; Circle 2026a; International Organization of Securities Commissions 2025). The oracle and proof-of-backing problem is therefore not merely technical; it is institutional and evidentiary.

Equally important, RWA market metrics are often conflated. Outstanding asset value is not the same as TVL. TVL is not the same as primary issuance. Minting and redemption activity are not the same as secondary-market liquidity. Transfer volume is not necessarily trading volume. A tokenized fund may have large assets outstanding while exhibiting minimal secondary trading; a DeFi protocol may show large TVL while revealing little about actual capital-market liquidity;

and a tokenized product may display frequent operational transfers without meaningful price-discovery depth (DefiLlama 2026; RWA.xyz 2026c; RWA.xyz 2026a; RWA.xyz 2026b; Mafrur 2025; Mafrur and Khadijah 2026; Mafrur 2026). For selection purposes, issuers should therefore treat *size*, *activity*, and *liquidity* as related but non-identical indicators.

The empirical landscape to date supports a cautious reading. Official and regulatory reviews consistently argue that tokenization remains nascent, uneven across asset classes, and deeply dependent on settlement assets, interoperability, legal certainty, and market-infrastructure readiness (Aldasoro et al. 2023; Financial Stability Board 2024; International Organization of Securities Commissions 2025; Organisation for Economic Co-operation and Development 2025; European Central Bank 2025b; Monetary Authority of Singapore 2024a). Academic and applied RWA literature reaches a similar conclusion: the most scalable products are usually those with comparatively standardized underlying assets, strong custodial arrangements, simple redemption logic, and institutionally credible wrappers (Baltais et al. 2024; J. Li 2025; Luo et al. 2026; Chemla and Tinn 2026; Vella et al. 2026).

3 Blockchain and issuance-stack heterogeneity

Blockchain networks differ not only in consensus and developer tooling, but also in privacy models, finality assumptions, permissioning support, native asset controls, and the extent to which regulated-asset functionality is built into the network, token standard, or external middleware. Ethereum-style environments tend to externalize regulated-asset logic into token standards and smart-contract frameworks such as ERC-1400, ERC-1450, ERC-1462, and ERC-3643, while newer initiatives such as ERC-7943 aim to generalize RWA compliance interfaces (Vogelsteller and Buterin 2015; Dossa et al. 2018; Marks et al. 2018; Kupriianov and Svirsky 2018; Lebrun et al. 2021; Lo Buglio et al. 2025; Tokeny 2023; Tokeny 2026). Stellar embeds different design choices through issuer flags, trustlines, anchor standards, and clawback-enabled regulated assets (Stellar Development Foundation 2026; Stellar Development Foundation 2025b; Stellar Development Foundation 2025a; Stellar Development Foundation 2025d; Stellar Development Foundation 2025c). Solana exposes another model through Token-2022 extensions such as transfer hooks, confidential transfers, transfer fees, pausable tokens, non-transferability, and permanent delegates (Solana Foundation 2026a; Solana Foundation 2026b; Solana Foundation 2024). Permissioned or privacy-oriented infrastructures such as Corda, Canton, Hyperledger Besu, and institutional Avalanche deployments emphasize selective data visibility, managed network participation, and custom governance or notary arrangements (Hearn and Brown 2019; R3 2024a; R3 2023; R3 2024b; Digital Asset 2026a; Digital Asset 2026b; Hyperledger 2019; Hyperledger Besu 2026; Ava Labs 2023; Ava Labs 2024; Ava Labs 2026).

This diversity implies that “blockchain selection” is partly misnamed. In practice, issuers choose among architecture families. Some prefer public chains with regulated token standards and external compliance layers; others use permissioned or semi-permissioned networks to embed privacy, governance, or participant admission more deeply; still others adopt hybrid arrangements where issuance, transfer control, and settlement span multiple networks or interoperating systems (Belchior et al. 2021; Monetary Authority of Singapore 2025c; SWIFT 2025; SWIFT 2026b; European Central Bank 2025a). The selection object is therefore the end-to-end oper-

ating model.

A second source of heterogeneity lies in institutional interfaces. For funds and securities, transfer agents, administrators, custodians, qualified custodians, trustees, and regulated trading venues often remain indispensable even when the token itself is on-chain (U.S. Securities and Exchange Commission 2025c; Investor.gov 2025; Securitize 2026; Apex Group 2026; BitGo 2026; BitGo 2025; Fireblocks 2025). Tokenization can alter workflows, timing, and reconciliation burdens, but it does not necessarily remove these institutions. Indeed, IOSCO emphasizes that tokenization often preserves the economic substance of the asset while changing the technological medium and process flow, which can introduce new actors such as token minters, wallet providers, or DLT platform operators (International Organization of Securities Commissions 2025).

A third heterogeneity concerns settlement assets. The ability to pair an RWA transfer with cash-like settlement or near-instant redemption materially affects usability. Some products rely on stablecoins, some on tokenized deposits or bank liabilities, some on conventional off-chain cash rails, and some on hybrid mechanisms. Official work by BIS, ECB, MAS, and SWIFT repeatedly highlights that credible settlement assets and interoperability with existing payment infrastructure remain central bottlenecks to scale (Committee on Payments and Market Infrastructures 2024; Bank for International Settlements 2025b; European Central Bank 2025b; European Central Bank 2025a; Monetary Authority of Singapore 2025d; SWIFT 2026a). Even where a tokenized asset is technically transferrable 24/7, investor utility depends on whether the payment leg, subscription leg, or redemption leg is equally credible and timely.

A fourth heterogeneity concerns access to liquidity and composability. Products such as BUIDL, FOBXX, OUSG, and USYC illustrate different choices about investor eligibility, mint and redeem windows, supported networks, and connection to DeFi or collateral workflows (U.S. Securities and Exchange Commission 2024; U.S. Securities and Exchange Commission 2025a; Securitize 2024b; Securitize 2024a; Franklin Templeton 2026; U.S. Securities and Exchange Commission 2025b; Ondo Finance 2026a; Ondo Finance 2026b; Circle 2026d; Circle 2026c). DeFi integration is possible, but it is mediated by wrappers, collateral rules, whitelists, oracle designs, and protocol risk parameters than implied by the chain itself (Aave Companies 2022; Adams et al. 2021; Centrifuge 2021; Centrifuge 2025; Lambur et al. 2018; Benligiray et al. 2020; Breidenbach et al. 2021; Schär 2021).

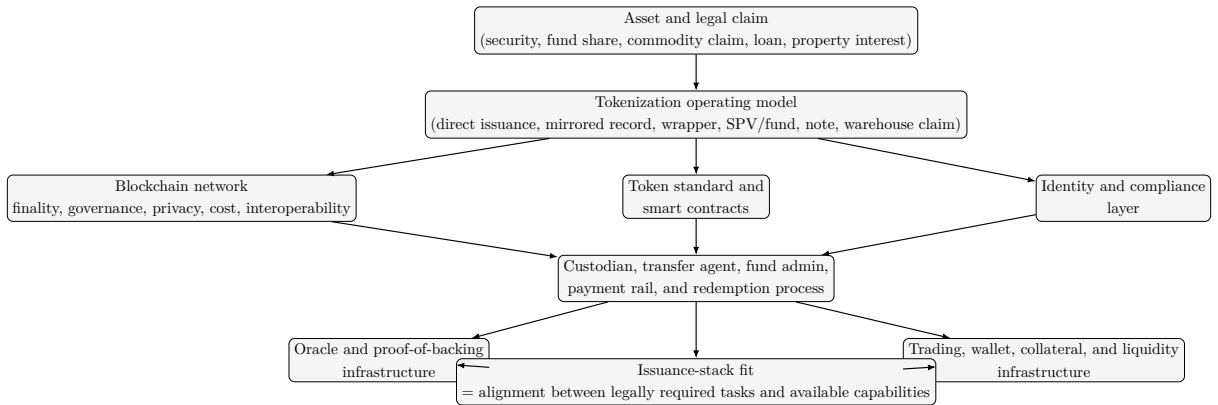


Figure 1: RWA issuance-stack architecture

4 Theoretical foundation

We used Task–Technology Fit as the principal theory for this review. In TTF terms, the *task* is issuing, managing, transferring, settling, redeeming, and governing tokenized RWAs. *Task requirements* are the legal, regulatory, operational, market, and technical conditions that must be satisfied. The *technology* is not merely the blockchain base layer, but the complete issuance stack. *Fit* is the degree to which the selected stack supports the required task performance under the relevant constraints (Goodhue and Thompson 1995; Zigurs and Buckland 1998; Dishaw and Strong 1999).

This theoretical framing is useful for three reasons. First, it directs attention away from decontextualized network traits and toward requirement satisfaction. High throughput or low fees may improve performance only if the required legal and operational controls are already feasible. Second, it explains why a public chain, a permissioned ledger, or a hybrid stack can each be appropriate depending on the task. Third, it clarifies why feasibility conditions are prior to preference-based optimization. A technology with poor task fit cannot be rescued by ecosystem size, and a technology with high generic popularity may still be infeasible for a regulated instrument.

Contingency theory and institutional perspectives add secondary value. Contingency theory supports the claim that there is no universally optimal network architecture because performance depends on the decision environment, including asset class, jurisdiction, participant type, privacy needs, and settlement structure (Donaldson 2001). Institutional theory helps explain why regulated incumbents often favor stacks that preserve recognized intermediaries, standardized control points, and familiar forms of authority, even where public-chain alternatives exist (DiMaggio and Powell 1983). The technology–organization–environment perspective is also relevant at the margin because organizational capability, supervisory climate, and ecosystem readiness shape what can actually be implemented (Tornatzky and Fleischer 1990). Nonetheless, TTF remains the clearest primary framework because the practical issuer question is fundamentally one of requirements matching.

5 Review methodology

This study uses a systematic review informed by PRISMA 2020 reporting principles, particularly those concerning search transparency, eligibility criteria, source selection, and structured synthesis. An integrative design was selected because the research question requires evidence from heterogeneous source types, including peer-reviewed studies, working papers, legal and regulatory materials, institutional reports, technical standards, issuer filings, governing documents, and official product documentation.

5.1 Source identification

The search covered Google Scholar, publisher databases, SSRN, arXiv, and the official websites or document repositories of the Bank for International Settlements, International Organization of Securities Commissions, Financial Stability Board, International Monetary Fund, Organ-

isation for Economic Co-operation and Development, World Bank, European Central Bank, Monetary Authority of Singapore, U.S. Securities and Exchange Commission, EDGAR, and EUR-Lex.

Technical and implementation evidence was obtained from official documentation published by Ethereum, Stellar, Solana, R3/Corda, Digital Asset/Canton, Avalanche, Hyperledger Besu, Chainlink, SWIFT, Tokeny, Securitize, Circle, Paxos, Franklin Templeton, Ondo Finance, RWA.xyz, and DefiLlama.

The principal review period covered materials published from 2015 to July 2026. Earlier sources were included selectively when required for foundational theory, review methodology, legal concepts, or established financial-market principles.

Search terms were iteratively combined around the following concepts: “tokenized real-world assets”, “RWA tokenization”, “tokenized Treasuries”, “tokenized funds”, “tokenized securities”, “security tokens”, “asset tokenization”, “blockchain network selection”, “transfer agent”, “administrative control”, “proof of reserve”, “delivery versus payment”, “settlement finality”, “privacy”, “interoperability”, and “tokenized settlement”.

Targeted searches also used relevant product, project, network, and standard names, including BUIDL, FOBXX, OUSG, USYC, PAXG, ERC-3643, ERC-1450, ERC-7943, Token-2022, SEP-24, Project Guardian, Corda, and Canton.

The search process initially identified approximately 320 potentially relevant records. The initial corpus included duplicate records, superseded versions, documents with insufficient relevance, and sources that did not provide enough legal, technical, operational, or market detail to support the analysis.

5.2 Eligibility and screening

Sources were retained when they satisfied the following criteria:

1. they were relevant to at least one RWA issuance, transfer, settlement, servicing, redemption, governance, or evidentiary task;
2. they provided evidence relating to at least one requirement or issuance-stack capability;
3. they were sufficiently authoritative for the type of claim being supported; and
4. they contained enough methodological, legal, technical, or operational detail to permit structured extraction.

Sources were excluded when they were duplicative, superseded by a more authoritative or current version, unrelated to the defined research questions, insufficiently detailed, primarily promotional without verifiable supporting evidence, or of unclear authorship or provenance.

Following deduplication, relevance screening, eligibility assessment, and source-quality appraisal, 158 unique sources were retained in the final evidence base. The retained corpus consisted of peer-reviewed research, academic working papers, legislation and regulatory materials, institutional reports, official technical standards and documentation, issuer filings and governing documents, and selected official product and implementation materials.

5.3 Source appraisal

Source authority was evaluated in relation to the type of claim being made. Legislation, regulatory materials, and official guidance were prioritized for jurisdiction-specific legal and regulatory claims. Peer-reviewed studies and institutional research were prioritized for theoretical, analytical, and market-level claims. Official standards and technical documentation were used to establish documented protocol and token-standard capabilities. Issuer filings, governing documents, and formal product disclosures were preferred for claims concerning holder rights, custody, eligibility, redemption, and product structure.

Commercial or vendor-produced materials were used primarily to document current implementation features. Where those materials made broader claims concerning legal certainty, security, market adoption, or operational effectiveness, they were triangulated against regulatory, institutional, academic, or other authoritative evidence.

5.4 Data extraction and synthesis

For each retained source, the review recorded, where applicable, the source type, asset class, jurisdictional scope, legal structure, requirement dimensions, capability locus, governance assumptions, settlement design, custody arrangements, administrative functions, privacy model, market infrastructure, proof-of-backing arrangements, and stated limitations.

Thematic coding followed an iterative abductive process informed by Task–Technology Fit theory. Evidence was organized into five principal clusters:

1. legal and regulatory requirements;
2. operational and lifecycle requirements;
3. technical and security requirements;
4. market and liquidity requirements; and
5. proof-of-backing and evidentiary requirements.

The final synthesis was used to construct the requirement taxonomy, capability taxonomy, mandatory feasibility gates, comparative selection criteria, issuer decision tree, and worked private-credit application.

6 Review findings

The review yields five principal findings.

First, there is no single “best” blockchain for RWA tokenization. Both academic and official sources consistently show that suitability is context-dependent and affected by network architecture, instrument type, participant restrictions, operational workflow, and market structure (Goodhue and Thompson 1995; Donaldson 2001; Aldasoro et al. 2023; International Organization of Securities Commissions 2025; Financial Stability Board 2024; Organisation for Economic

Table 1: Aggregate source-identification and selection results

Review stage	Description	Number
Initial identification	Potentially relevant academic, regulatory, institutional, technical, issuer, and market-infrastructure records identified through database and targeted website searches.	Approx. 320
Deduplication and screening	Duplicate records, superseded versions, insufficiently relevant sources, promotional or unverifiable material, and documents lacking sufficient analytical detail were removed.	Not recorded separately
Final evidence base	Unique sources retained for structured extraction, thematic coding, and framework development.	158

Note: The initial identification count is approximate because stage-specific search exports were not preserved. The final count represents the unique sources retained and used in the synthesis.

Co-operation and Development 2025; Agur and Copestake 2025). This is not a superficial preference statement. It is an empirical and analytical consequence of heterogeneity in tasks and constraints.

Second, legal and institutional feasibility dominates purely technical optimization. If the legally authoritative ownership record cannot be aligned with the token model, if transfer-agent or fund-administration support is unavailable, if investor eligibility cannot be enforced, if recovery or court-ordered transfer cannot be operationalized, or if redemption cannot be executed reliably, then a candidate stack is infeasible regardless of throughput or fee advantages (Konashevych 2020; World Bank 2020; U.S. Securities and Exchange Commission 2026; U.S. Securities and Exchange Commission 2025c; Investor.gov 2025; International Organization of Securities Commissions 2025). This finding directly motivates a mandatory feasibility screen prior to any comparative assessment.

Third, asset classes differ in what they require from tokenization. Tokenized Treasuries and money market funds typically emphasize subscription and redemption control, investor onboarding, NAV updates, yield distribution, and integration with stablecoin or cash settlement. Commodity tokens emphasize custody, auditability, serial-number or bar-level traceability where applicable, and redemption enforceability. Private-credit structures emphasize confidentiality, servicing, covenant and waterfall administration, and restricted transfer. Tokenized equities and regulated fund securities emphasize authoritative shareholder records, corporate actions, proxy voting, transfer-agent control, and market-infrastructure integration (Monetary Authority of Singapore 2025b; Monetary Authority of Singapore 2025a; Franklin Templeton 2026; Ondo Finance 2026b; Circle 2026c; Paxos 2025; Paxos 2026; U.S. Securities and Exchange Commission 2025a). Table 2 summarises the RWA structures and requirements.

Fourth, many capabilities that market participants colloquially attribute to a blockchain are not native to the base chain at all. Network-level functionality generally covers consensus, ledger ordering, finality model, public visibility, and general smart-contract execution. Compliance features such as allowlists, partitions, recovery powers, or transfer hooks are usually implemented through token standards or application contracts; identity is often supplied through middleware or issuer-controlled registries; proof of reserves depends on external attestations and oracle design; custody and transfer-agent functions depend on licensed institutions; and settlement quality depends on the credibility of the cash leg and associated market infrastructure (Lebrun et al. 2021; Marks et al. 2018; Stellar Development Foundation 2026; Stellar De-

velopment Foundation 2025b; Solana Foundation 2026a; Tokeny 2023; Chainlink 2025b; Circle 2026a; BitGo 2026; Securitize 2026). Network selection therefore cannot be assessed correctly without capability decomposition.

Fifth, adoption and liquidity metrics must be interpreted carefully. Outstanding value may indicate product uptake or reserve size, but says little about secondary liquidity by itself. TVL is a protocol-centric measure, not a direct measure of transferable, tradeable, or redeemable RWA value. Transfer volume can capture operational movements, subscriptions, redemptions, or internal routing. Secondary-market liquidity requires evidence of counterparties, venues, bid–ask depth, turnover, and redemption pathways. On-chain observability, meanwhile, does not prove reserve quality, legal title, or encumbrance status (DefiLlama 2026; RWA.xyz 2026c; RWA.xyz 2026a; Mafrur 2025; Mafrur and Khadijah 2026; Mafrur 2026; Chainlink 2025a; International Organization of Securities Commissions 2025).

Table 2: RWA structures and priority requirements

Asset class	Typical tokenized claim or structure	Priority requirements
Treasuries and cash-management products	Direct sovereign security; regulated fund share; note or other beneficial exposure	Clear investor rights; eligibility controls; NAV or price determination; custody; cash-leg settlement; subscriptions and redemptions; asset reconciliation; collateral eligibility where relevant.
Bonds	Natively issued digital bond; tokenized conventional bond; note or beneficial interest	Authoritative ownership record; registrar or CSD integration where applicable; delivery-versus-payment; settlement finality; coupon and principal servicing; transfer restrictions; default and corporate-action processing.
Private credit	Fund or SPV interest; participation; receivable-backed claim; lender note	Borrower and collateral due diligence; servicing and payment waterfalls; covenant monitoring; confidentiality; transfer controls; valuation; default and workout procedures; collateral and cash-flow reporting.
Investment funds	On-chain share class; tokenized fund share or unit; feeder or nominee interest	Transfer-agent and administrator integration; investor eligibility; NAV calculation; subscriptions and redemptions; distributions; custody and cash management; reconciliation with the official register.
Equities	Issuer-sponsored digital share; tokenized conventional share; depositary, nominee, or contractual exposure	Legal characterization of holder rights; authoritative shareholder record; transfer-agent or registrar support; voting, dividends, and corporate actions; trading and transfer restrictions; custody and investor protection.
Commodities	Allocated custody-backed claim; warehouse receipt; fund, note, or contractual exposure	Clear title and claim priority; qualified custody or warehouse controls; inventory reconciliation; audit or attestation; insurance and asset-quality verification; sanctions screening; redemption and delivery terms.
Real estate	SPV equity or debt; fund or trust interest; revenue-sharing or secured-credit claim	Separation between the token and official land title; SPV governance; valuation; income and expense allocation; servicing; investor rights; transfer restrictions; enforcement and exit mechanisms; realistic secondary-market assumptions.

Note: The structures and requirements are jurisdiction-dependent. A token may evidence direct legal ownership, beneficial ownership, or only a contractual claim against an issuer or intermediary. Blockchain records should therefore be reconciled with the legally authoritative register and the instrument’s governing documents.

7 Requirements taxonomy

For analytical clarity, we grouped requirements into five clusters: legal and regulatory requirements; operational and lifecycle requirements; technical and security requirements; market and liquidity requirements; and proof-of-backing and evidentiary requirements. These dimensions jointly define the *task profile* against which issuance-stack fit must be evaluated.

Legal and regulatory requirements

Legal and regulatory requirements determine whether the tokenized instrument is recognizable, enforceable, and distributable in the intended jurisdiction. They include the legal nature of the tokenized claim, compatibility with the authoritative ownership record, investor eligibility rules, transfer restrictions, recovery and forced-transfer powers, reporting obligations, custody rules, and privacy or disclosure constraints (World Bank 2020; Konashevych 2020; International Organization of Securities Commissions 2025; U.S. Securities and Exchange Commission 2026; European Union 2022; European Union 2023). For many securities and funds, these are not optional enhancements. They are feasibility conditions. Table 3 summarises the core legal and regulatory requirements for RWA tokenization.

Table 3: Core legal and regulatory requirements for RWA tokenization

Requirement	Typical applicability	Regulatory significance and design implication
Legal characterization and enforceable holder rights	Core	Defines the instrument, issuer or obligor, governing law, holder rights, claim priority, dispute process, and insolvency treatment. The token must represent an enforceable legal interest
Authoritative record and reconciliation	Core where the token records ownership or entitlement	Determines whether the distributed ledger is the official record, part of that record, or only a supplemental record. Conflicting records, correction procedures, and reconciliation responsibilities must be defined.
Regulated roles, permissions, and licences	Activity- and jurisdiction-dependent	Issuance, distribution, trading, custody, brokerage, transfer agency, fund administration, and settlement may require licensed entities or approved market infrastructure.
Offering, investor eligibility, and financial-crime controls	Commonly applicable	Supports compliance with offering restrictions, investor-category rules, customer due diligence, sanctions controls, and geographic limitations. Controls may operate on-chain, off-chain, or through both layers.
Transfer and holding restrictions	Conditional	Required where only specified persons or jurisdictions may acquire, hold, or receive the instrument. The system must also accommodate legally valid exceptions, corrections, and authorized transfers.
Administrative intervention and recovery	Conditional; explicit legal basis required	Freeze, forced transfer, error correction, court-order execution, and key-loss recovery may be necessary in some structures. Authorized actors, triggering conditions, procedural safeguards, and auditability must be defined.
Custody, control, and client-asset segregation	Core when assets or keys are held for clients	Requires clear control arrangements, safekeeping, segregation, recovery procedures, liability allocation, and protection of client assets in insolvency or operational failure.
Disclosure, reporting, and record retention	Core for regulated instruments and activities	Disclosures should explain holder rights, fees, technology risks, custody, transfer limitations, redemption, and conflicts of interest. On-chain traceability does not by itself satisfy regulatory reporting or recordkeeping duties.
Privacy and data governance	Applicable where personal or confidential information is processed	Requires data minimization, controlled access, lawful processing, retention and correction procedures, and appropriate separation of on-chain records from sensitive off-chain information.
Settlement, redemption, and wind-down certainty	Product- and infrastructure-dependent; often critical	Defines legal settlement finality, the cash leg, failed-settlement handling, redemption rights, asset return, continuity arrangements, and migration or wind-down procedures.

Note: “Core” denotes a condition generally essential to a legally viable issuance stack, not a requirement imposed identically by every jurisdiction. Applicability depends on the instrument’s legal characterization, jurisdiction, investor type, distribution model, and regulated activities (U.S. Securities and Exchange Commission 2026; International Organization of Securities Commissions 2025).

Operational and lifecycle requirements

Operational requirements concern how the asset behaves after issuance. The review identifies recurring requirements related to subscription and issuance workflows, corporate actions, income distributions, pricing and valuation, settlement timing, redemptions, exception handling,

servicing, and off-chain administration (Monetary Authority of Singapore 2025b; Monetary Authority of Singapore 2025a; U.S. Securities and Exchange Commission 2025b; U.S. Securities and Exchange Commission 2025a; Ondo Finance 2026b; Circle 2026c). These are the areas where tokenization most often encounters business-process friction. Table 4 summarises operational and lifecycle requirements for RWA issuance stacks.

Table 4: Operational and lifecycle requirements for RWA issuance stacks

Lifecycle function	Typical applicability	Operational requirement
Issuance, subscription, and onboarding	Core for primary issuance and direct distribution	Coordinate investor verification, eligibility approval, order acceptance, pricing, receipt of funds, token minting or allocation, and confirmation of the investor’s legal entitlement.
Valuation, pricing, and NAV	Product-dependent	Establish authoritative data sources, valuation frequency, approval responsibility, correction procedures, and publication controls. Particularly important for funds, pooled vehicles, private assets, and collateralized products.
Asset servicing and distributions	Applicable to income-bearing instruments	Process coupons, dividends, interest, principal repayments, fees, tax withholding, and other distributions. Calculation and payment may require off-chain agents and payment infrastructure.
Corporate actions and governance	Material for equities, debt, and some fund structures	Support voting, consent solicitations, calls, conversions, restructurings, splits, mergers, and other holder events, with holdings reconciled to the authoritative record at the relevant record date.
Settlement and cash-leg coordination	Core for transactions involving payment	Define delivery-versus-payment arrangements, settlement timing, permitted settlement assets, legal finality, failed-settlement procedures, and the relationship between on-chain transfers and external payment systems.
Position, supply, and record reconciliation	Core where multiple records or intermediaries exist	Reconcile token supply and balances with issuer, registrar, custodian, administrator, CSD, reserve, or underlying-asset records; investigate and resolve discrepancies promptly.
Redemption, cancellation, and asset return	Core where redemption or conversion rights exist	Verify holder eligibility, calculate the redemption amount, arrange payment or asset delivery, burn or immobilize tokens, update official records, and manage liquidity, notice periods, fees, and suspensions.
Exceptions, corrections, and recovery	Core	Address failed or erroneous transactions, duplicate issuance, lost keys, sanctions or court orders, disputed ownership, smart-contract incidents, and servicing errors through authorized and auditable procedures.
Credit and receivable servicing	Core for credit products	Manage payment collection, allocation waterfalls, covenant monitoring, arrears, collateral reporting, defaults, enforcement, restructurings, and investor reporting. These functions commonly depend on off-chain servicers.
Operational resilience and wind-down	Core	Provide business-continuity arrangements, access-control and key-management procedures, system recovery, contract upgrades, data portability, migration, and an orderly wind-down process if a provider or network becomes unavailable.

Note: Applicability depends on the asset, legal structure, jurisdiction, distribution model, and supporting institutions. Completion of an on-chain transaction does not necessarily establish legal settlement, update the authoritative ownership record, or complete the cash leg. Lifecycle functions should therefore be assessed across the full issuance stack rather than the blockchain alone (International Organization of Securities Commissions 2025; Committee on Payments and Market Infrastructures 2024; U.S. Securities and Exchange Commission 2026).

Technical and security requirements

Technical requirements include security, resilience, governance, finality, scalability, privacy behavior, upgradeability, interoperability, and the ability to implement transfer and identity controls at the correct layer (Croman et al. 2016; Garay et al. 2015; Lewenberg et al. 2015; Belchior et al. 2021; International Organization of Securities Commissions 2025; Financial Stability Board 2024). RWA use cases elevate finality and governance because legal settlement and roll-back expectations are not identical to DeFi experimentation. Table 5 summarises technical, cybersecurity, and infrastructure requirements.

Table 5: Technical, cybersecurity, and infrastructure requirements

Requirement	Typical applicability	Design implication
Protocol and network security	Core	Assess consensus security, validator or operator concentration, transaction censorship, chain reorganizations, denial-of-service exposure, network maturity, and the consequences of protocol failure or disruption.
Smart-contract assurance	Core where programmable contracts are used	Require secure design, testing, independent review, controlled deployment, runtime monitoring, and procedures for vulnerabilities. Contract failure may affect token control, transfers, servicing, or settlement even where the underlying legal claim survives.
Key management and access control	Core	Define secure key generation, storage, authorization, signing, backup, recovery, rotation, revocation, and compromise response for issuers, custodians, administrators, and other privileged actors.
Operational resilience and recoverability	Core	Provide redundancy, monitoring, incident response, disaster recovery, business continuity, data recovery, and migration arrangements across the blockchain and all dependent service providers.
Transaction and settlement finality	Core where transfers settle legal or financial obligations	Distinguish protocol-level confirmation from legal settlement finality. Specify required confirmation thresholds, treatment of reorganizations, cash-leg completion, record reconciliation, and failed-settlement procedures.
Governance, upgrades, and change control	Core	Identify who can modify contracts, parameters, permissions, or network software; require authorization controls, testing, audit trails, notice periods, emergency procedures, and contingency plans for contentious network changes.
Data, oracle, and proof integrity	Core where external information affects token functions	Verify the source, timeliness, accuracy, authorization, redundancy, and failure handling of prices, NAVs, interest rates, identity status, reserve data, corporate actions, and other off-chain inputs.
Privacy and selective disclosure	Conditional	Use data minimization, permissioned access, encryption or privacy-preserving methods where personal, transactional, or commercially sensitive information is processed. Sensitive data should generally remain off-chain where appropriate.
Capacity, performance, and transaction cost	Context-dependent	Set minimum thresholds for throughput, latency, predictable fees, congestion handling, and service availability. Requirements depend on transaction frequency, investor scale, lifecycle activity, and product economics.
Interoperability and external integration	Conditional	Assess integration with payment systems, custodians, registrars, identity providers, legacy infrastructure, and other ledgers. Additional connections may expand distribution but also increase attack surfaces, reconciliation needs, and dependency risk.
Technology-provider and dependency risk	Core where critical third parties are used	Evaluate reliance on node providers, cloud services, custodians, bridges, oracle operators, wallet providers, software libraries, and proprietary platforms, including concentration, substitution, portability, and exit risk.

Note: Technical suitability should be evaluated across the complete issuance stack, not solely at the base-blockchain level. On-chain confirmation does not by itself establish legal settlement finality, complete the payment leg, or ensure consistency with the authoritative ownership record. Interoperability may improve reach and integration but can also introduce additional security, governance, and operational dependencies (Financial Stability Board 2024; International Organization of Securities Commissions 2025; Committee on Payments and Market Infrastructures 2024).

Market and liquidity requirements

Market requirements concern the ability to distribute, trade, collateralize, and settle the tokenized instrument in economically meaningful ways. These requirements span wallet and custody support, stablecoin and payment availability, venue access, collateral eligibility, market-making potential, and interoperability with existing financial infrastructure (Amihud and Mendelson 1986; Campbell and Shiller 1988; Financial Stability Board 2024; Monetary Authority of Singapore 2025c; SWIFT 2026a; European Central Bank 2025b). The key lesson is that “being on-chain” is not itself liquidity. Table 6 summarises market access, settlement, and liquidity requirements.

Table 6: Market access, settlement, and liquidity requirements

Market function	Typical applicability	Design implication
Investor access, wallets, and custody	Core for the intended distribution model	Ensure that eligible investors and intermediaries can acquire, hold, transfer, and recover access to the token through compatible wallets, custodians, or intermediated account structures. Broad wallet availability does not by itself establish investor demand or liquidity.
Settlement-asset and payment access	Core for transactions against payment	Specify the permitted cash leg, such as central-bank money, commercial-bank money, tokenized deposits, stablecoins, or conventional payment rails. Assess credit, liquidity, redemption, legal-finality, availability, and interoperability risks.
Primary issuance and distribution	Core for new issuance	Define subscription channels, investor onboarding, allocation, pricing, payment, issuance limits, placement restrictions, and the roles of distributors or intermediaries.
Secondary-market access	Conditional	Determine whether transfers may occur bilaterally, through an approved venue, dealer network, alternative trading system, exchange, or not at all. Venue access must be legally permitted and operationally compatible with transfer and investor restrictions.
Market depth and price discovery	Conditional but essential where liquidity is claimed	Evaluate the number and diversity of participants, trading frequency, turnover, bid-ask spreads, executable depth, valuation sources, and market-making arrangements. Issuance size, token-holder count, TVL, or on-chain transfer volume should not be used alone as evidence of liquidity.
Liquidity management and redemption capacity	Core for redeemable or liquidity-transforming products	Align redemption terms with the liquidity of the underlying assets and available cash buffers. Address notice periods, gates, suspensions, settlement delays, stressed redemptions, and the risk that token transferability exceeds underlying-asset liquidity.
Collateral, financing, and repo eligibility	Conditional	Confirm legal enforceability, valuation and margin processes, custody and control arrangements, settlement compatibility, eligible counterparties, and acceptance by the relevant lender, market infrastructure, or central bank. Tokenization alone does not confer collateral eligibility.
Market data and reference information	Product-dependent	Provide reliable prices, NAVs, rates, corporate actions, reserve information, and transaction data where required. Define authoritative sources, update frequency, licensing, validation, correction, and fallback procedures.
Programmable-finance and DeFi integration	Optional and risk-dependent	Assess legal permissibility, investor eligibility, smart-contract and oracle risk, liquidity quality, governance, sanctions controls, leverage, rehypothecation, and the consequences of interaction with third-party protocols.
Cross-platform distribution and interoperability	Conditional	May expand investor and venue access, but can fragment liquidity and introduce bridge, custody, reconciliation, governance, and operational dependencies. Use only where the additional reach justifies the added complexity and risk.

Note: Market access and market liquidity are distinct. A token may be transferable through compatible wallets without having an active, legally accessible, or sufficiently deep secondary market. Similarly, issuance volume, assets under management, total value locked, holder count, and on-chain transfer activity do not independently demonstrate executable liquidity. Settlement-asset, collateral, and DeFi capabilities should be assessed across the complete issuance and market-infrastructure stack (Organisation for Economic Co-operation and Development 2025; International Organization of Securities Commissions 2025; Committee on Payments and Market Infrastructures 2024; Financial Stability Board 2024).

Reserve assurance, proof-of-backing, and evidentiary requirements

Reserve-backed and off-chain-linked tokens require more than an on-chain reserve balance. A credible assurance framework must establish the existence, ownership or control, quality, valuation, custody, segregation, and availability of the referenced assets; reconcile those assets with the token supply and other outstanding claims; and explain the holder’s redemption and enforcement rights. The frequency, independence, scope, and methodology of the evidence are also material.

Accordingly, proof of backing should be understood as a layered evidentiary framework. Automated reserve feeds may improve the timeliness and usability of verified information, including by supporting minting controls or circuit breakers, but they do not independently establish legal title, bankruptcy remoteness, absence of encumbrances, asset quality, or enforceable redemption rights (Chainlink 2025b; Chainlink 2025a; Circle 2026a; Paxos 2025; International Organization of Securities Commissions 2025). Table 7 summarises reserve-assurance and proof-of-backing requirements.

Table 7: Reserve-assurance and proof-of-backing requirements

Evidentiary requirement	Typical applicability	Required evidence or control
Existence and identification of backing assets	Core for asset-backed products	Custodian, bank, trustee, administrator, warehouse, or issuer records that identify the assets held, their location, relevant accounts, and the date to which the evidence relates.
Legal title, control, and claim priority	Core	Governing documents and legal analysis establishing who owns or controls the backing assets, the rights represented by the token, the holder’s priority, and treatment in issuer, custodian, or intermediary insolvency.
Segregation and custody arrangements	Core where assets are held by an issuer or third party	Evidence that backing assets are held in the promised accounts or custody structure, separated where required from proprietary assets, and subject to defined access, withdrawal, and substitution controls.
Encumbrances and competing claims	Core	Disclosure and verification of pledges, liens, securities interests, rehypothecation, securities lending, guarantees, or other arrangements that could restrict the availability of the backing assets.
Asset eligibility, quality, and liquidity	Product-dependent but often material	Evidence that reserve assets satisfy the instrument’s eligibility criteria, including maturity, credit quality, concentration, market liquidity, insurance, physical quality, or other product-specific conditions.
Valuation and measurement	Core where backing is measured by value	Documented valuation methods, authoritative price or NAV sources, valuation times, accrued income, foreign-exchange treatment, haircuts, stale-price controls, and correction procedures.
Token-supply and reserve reconciliation	Core	Regular reconciliation of issued, circulating, locked, burned, and pending tokens against eligible backing assets and other outstanding claims, including investigation and correction of discrepancies.
Minting, burning, and reserve-change controls	Core where token supply is issuer-controlled	Authorization and audit trails showing that minting follows valid receipt or allocation of backing assets and that redemptions, cancellations, and reserve releases produce corresponding supply changes.
Redemption rights and operational capacity	Core where redemption is promised	Legally enforceable terms specifying eligible redeemers, redemption assets, minimum amounts, fees, notice periods, processing times, payment or delivery arrangements, suspensions, and responsible counterparties.
Evidence frequency, scope, and independence	Core	Clear identification of whether evidence is real-time, periodic, or point-in-time; whether it is an audit, assurance engagement, attestation, custodian report, or management disclosure; and which assets, liabilities, entities, and chains are included or excluded.
Data-feed and oracle integrity	Applicable where evidence is transmitted on-chain	Transparent data provenance, authenticated sources, update thresholds, redundancy, monitoring, dispute procedures, and fallback controls when data are stale, unavailable, inconsistent, or disputed.
Exceptions, shortfalls, and failure treatment	Core	Predefined procedures for reserve shortfalls, delayed reporting, custody failure, valuation disputes, oracle failure, de-pegging, redemption suspension, unauthorized minting, and orderly wind-down or migration.

Note: An attestation, audit report, custodian statement, or on-chain proof-of-reserve feed provides assurance only within its stated scope and measurement period. Unless expressly covered, it should not be interpreted as proof of legal ownership, absence of encumbrances, completeness of liabilities, asset liquidity, bankruptcy remoteness, or unconditional redeemability.

8 Issuance-stack capability taxonomy

RWA functionality is distributed across four loci: the base ledger, the token and application layer, middleware and data services, and institutional and legal infrastructure. Distinguishing these loci prevents category errors. A blockchain does not itself provide KYC, legal title, custody, reserve assurance, fund administration, or market liquidity. It provides technical functions that may be combined with smart contracts, identity systems, oracles, payment infrastructure, and regulated institutions (Lebrun et al. 2021; Tokeny 2023; Stellar Development Foundation 2025d; Solana Foundation 2026b; Chainlink 2025b; BitGo 2026). Table 8 summarises capability loci within an RWA issuance stack.

Table 8: Capability loci within an RWA issuance stack

Capability	Primary locus	Operational role
Consensus, ordering, and state finality	Base ledger	Validates and orders transactions and commits ledger state; does not alone establish legal settlement finality.
Access and transaction visibility	Ledger and application	Controls network participation and data exposure. Confidentiality may require permissioning or application-level privacy controls.
Token issuance and transfer	Token or smart-contract layer	Provides balances, minting, burning, and transfer functions.
Transfer compliance and intervention	Token, identity, and institutional layers	Applies eligibility rules, allowlists, pauses, freezes, recovery, or forced transfers where legally authorized.
Identity and investor eligibility	Middleware and institution	Performs KYC, sanctions screening, and investor classification, then supplies verified status to the token system.
External data and reserve evidence	Oracle and institution	Supplies prices, NAVs, reserve data, rates, and corporate-action information from defined external sources.
Custody and key control	Institution and technology provider	Safeguards assets and keys and provides authorization, segregation, recovery, and incident-response controls.
Official records and lifecycle administration	Legal and institutional layer	Maintains authoritative registers, transfer agency, fund administration, asset servicing, and reconciliation.
Cash settlement and redemption	Payment and institutional infrastructure	Coordinates the cash leg, delivery-versus-payment, subscriptions, redemptions, and token cancellation.
Trading and liquidity	Market infrastructure	Provides permitted venues, dealers, market makers, or protocol access; token issuance alone does not create liquidity.

This taxonomy also qualifies the public-versus-permissioned distinction. Public chains can support regulated assets through application-level identity, transfer, and administrative controls. Permissioned networks can restrict participation and data visibility more directly, but still depend on legal documentation, custody, administration, payment, and market infrastructure. The two models therefore allocate institutional and technical control differently (Hearn and Brown 2019; R3 2024a; Digital Asset 2026a; Monetary Authority of Singapore 2025c; International Organization of Securities Commissions 2025).

Table 9: Requirements–capability matching matrix

Requirement	Delivery locus	Mechanism	Decision consequence
Legal title and holder rights	Legal and institutional	Governing documents, official register, trust, fund, or SPV structure	Exclude stacks that cannot align token records with enforceable rights.
Investor eligibility and transfer controls	Identity, token, and institution	KYC status, credentials, allowlists, and transfer validation	Mandatory gate where holding or distribution is restricted.
Administrative intervention	Token, contract, and institution	Pause, freeze, correction, recovery, or forced transfer under defined authority	Required only where supported by law and the instrument terms.
Privacy and data governance	Ledger, application, and institution	Restricted visibility, off-chain sensitive data, and selective disclosure	Must satisfy applicable confidentiality and data-protection obligations.
Settlement finality	Ledger, payment, and legal layers	Protocol finality, cash-leg completion, DvP, and settlement rules	Assess end-to-end settlement, not blockchain confirmation alone.
Proof of backing	Institution and oracle	Custody records, assurance, reserve reconciliation, and data feeds	On-chain reserve data is insufficient without legal and institutional evidence.
Lifecycle administration	Institution and contract	Official register, NAV, distributions, voting, and corporate actions	Requires integration with the authoritative administrator or recordkeeper.
Secondary liquidity	Market infrastructure	Approved venues, dealers, market makers, permitted protocols, or redemption	Token existence and transferability do not establish market liquidity.

9 Requirements-based decision framework

The proposed framework is deliberately two-stage. Stage 1 performs mandatory feasibility screening. Stage 2 conducts comparative assessment among only those alternatives that pass Stage 1. This sequencing is central to the TTF logic: technologies that fail essential task requirements should not enter preference-based optimization. Table 9 summarises requirements–capability matching matrix.

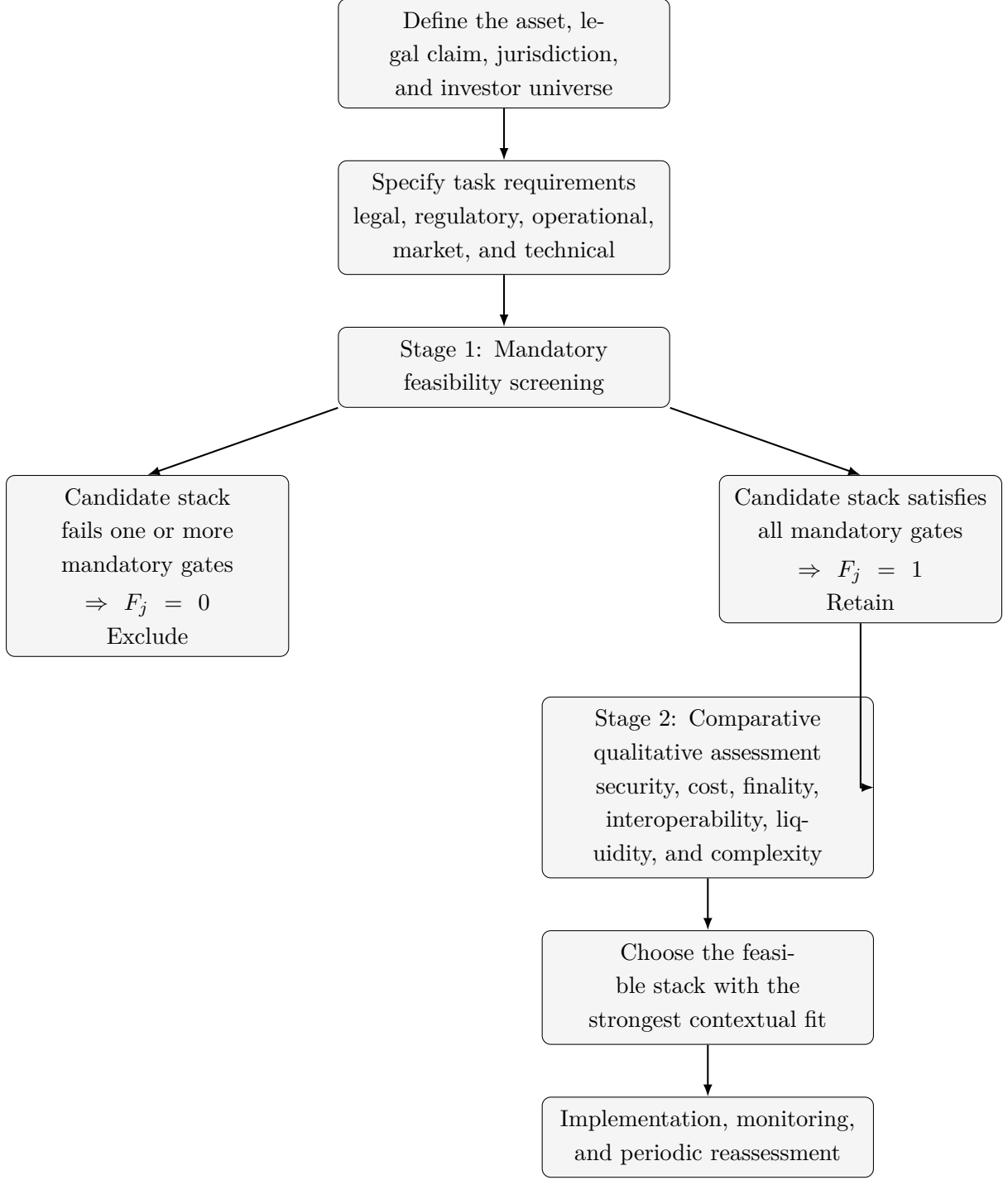


Figure 2: RWA network-asset fit framework

Formally, for candidate issuance stack j , feasibility is represented as:

$$F_j = \prod_{m=1}^M I(C_{jm} \geq R_m)$$

where C_{jm} denotes the capability level of candidate j on mandatory criterion m , R_m denotes the minimum required level for that criterion, and $I(\cdot)$ is an indicator function equal to 1 if the

condition is satisfied and 0 otherwise. The product structure captures conjunctive feasibility: if any mandatory criterion fails, then $F_j = 0$. In practical terms, a stack that cannot support legally required transfer restrictions, authoritative record alignment, or redemption cannot be justified merely because it offers lower fees, higher throughput, or a larger user base.

10 Mandatory feasibility gates

Stage 1 determines whether a candidate issuance stack can satisfy the minimum legal, regulatory, operational, and technical conditions of the proposed tokenized asset. The relevant gates must first be defined for the specific instrument, jurisdiction, investor population, and distribution model. A criterion is therefore mandatory only when its failure would make the proposed operating model unlawful, unenforceable, or operationally unworkable.

Table 10: Stage 1 mandatory feasibility gates

Gate	Pass condition	Failure consequence
Legal validity and enforceability	The token represents a legally recognized and enforceable instrument, entitlement, or contractual claim, with defined governing law, holder rights, claim priority, and insolvency treatment.	Restructure the instrument or discontinue tokenization.
Authoritative record alignment	The relationship between the token ledger and the legally controlling ownership or entitlement record is defined, including reconciliation, correction, and conflict-resolution procedures.	Exclude or redesign the recordkeeping model.
Regulatory permissions and institutional roles	The issuer, distributor, custodian, venue, administrator, transfer agent, and other service providers hold the permissions required for their activities.	Exclude the stack or replace the relevant provider.
Investor and transfer compliance	Where distribution or holding is restricted, the stack can enforce investor eligibility, jurisdictional limits, sanctions controls, and permitted transfers throughout the holding period.	Exclude the stack from the restricted distribution model.
Administrative controls	Where legally or contractually required, authorized parties can pause, freeze, correct, recover, or force transfers under defined governance and audit procedures.	Exclude or modify the token-control design.
Custody and lifecycle administration	Acceptable custody, key control, official recordkeeping, reconciliation, asset servicing, corporate actions, and investor communication arrangements are available.	Exclude the stack or redesign its institutional support.
Privacy, reporting, and data governance	The system can satisfy applicable confidentiality, personal-data, record-retention, disclosure, audit, and supervisory-reporting obligations.	Exclude or redesign the data architecture.
Settlement and payment	The asset and cash legs can be completed with the required operational and legal finality, including failure handling and reconciliation with external payment or settlement systems.	Exclude the proposed settlement arrangement.
Redemption and asset return	Where redemption is promised, eligible holders can obtain the specified cash, security, commodity, or other entitlement under enforceable and operationally credible terms.	The product cannot be marketed as redeemable.
Backing and reserve assurance	For asset-backed products, evidence supports asset existence, ownership or control, segregation, valuation, encumbrance status, reserve reconciliation, and token-supply integrity.	Exclude the structure or remove the backing claim.
Operational resilience and exit	The stack provides incident response, key recovery, continuity, data portability, provider substitution, migration, and orderly wind-down arrangements.	Exclude the stack if the deficiency cannot be remediated.

Note: Administrative intervention, investor restrictions, privacy controls, redemption, and proof of backing are not universal requirements. They become mandatory gates when required by the instrument’s legal terms, regulatory classification, jurisdiction, or operating model. Protocol confirmation does not by itself establish legal settlement finality, complete the payment leg, or update an external authoritative register (International Organization of Securities Commissions 2025; Committee on Payments and Market Infrastructures 2024; U.S. Securities and Exchange Commission 2026).

Candidates that satisfy all applicable gates proceed to Stage 2. Transaction fees, throughput, latency, interoperability, ecosystem reach, developer support, and liquidity infrastructure are normally comparative criteria. However, any such criterion becomes a feasibility gate when a defined minimum threshold is essential to the product. For example, transaction cost becomes mandatory if it makes subscriptions or servicing economically unviable, while latency becomes

mandatory if the settlement model requires completion within a specified period.

Accordingly, a candidate with $F_j = 0$ should be excluded even when it offers lower fees, higher throughput, or greater ecosystem visibility. In Task–Technology Fit terms, the candidate does not satisfy the minimum task conditions and should not receive compensating credit for superior preference-based attributes.

11 Comparative selection criteria

Stage 2 compares only candidates that satisfy all applicable feasibility gates. Its purpose is not to produce universal rankings or unsupported numerical weights, but to identify the strongest contextual fit within the feasible choice set (Goodhue and Thompson 1995; Dishaw and Strong 1999; Agur and Copestake 2025). Table 11 summarizes the principal comparative criteria.

Table 11: Stage 2 comparative selection criteria

Criterion	Assessment focus	Comparative question
Security and resilience	Protocol, contracts, keys, infrastructure, and recovery	Which feasible stack provides the strongest end-to-end security and operational resilience for the proposed use case?
Finality and settlement performance	Confirmation model, legal finality, latency, and failure handling	Which stack best satisfies the required settlement confidence and timing?
Total operating cost	Issuance, transfer, data, compliance, custody, and administration	Which stack provides the most sustainable lifecycle economics, including congestion and third-party service costs?
Capacity and predictability	Throughput, latency, availability, and fee stability	Which stack can support expected issuance, trading, and servicing activity under normal and stressed conditions?
Payment and settlement integration	Cash-leg assets, payment rails, DvP, and redemption	Which stack integrates most effectively with acceptable payment and settlement arrangements?
Custody and distribution coverage	Custodians, wallets, intermediaries, and investor access	Which stack best supports the target investors and distribution model?
Institutional and life-cycle integration	Registrars, administrators, venues, reporting, and asset servicing	Which stack requires the least complex integration with the institutions needed throughout the asset lifecycle?
Market access and liquidity infrastructure	Venues, dealers, market makers, collateral channels, and redemption	Which stack provides the most credible pathways to market access and liquidity without assuming that token issuance itself creates liquidity?
Interoperability and composability	External systems, other ledgers, applications, and programmable finance	Which stack delivers useful integration while limiting bridge, dependency, fragmentation, and governance risks?
Governance and operational maturity	Upgrade control, tooling, documentation, monitoring, and provider depth	Which stack offers the most credible long-term governance, implementation, maintenance, and support environment?

The criteria are intentionally qualitative and context-dependent. They should be assessed using documented evidence and explicit product priorities rather than fixed cross-case weights. For example, a tokenized Treasury or cash-management product may emphasize payment integration, custody coverage, and collateral connectivity, whereas a private-credit product may place greater weight on confidentiality, servicing integration, and administrative control. An issuer may also accept higher implementation complexity where it produces material distribution or programmability benefits.

A criterion that has already been treated as a mandatory gate should not be scored again merely because it is legally required. Stage 2 instead compares the relative quality of feasible implementations. For example, all retained stacks must meet the required security threshold, but they may still differ in resilience, governance, recovery procedures, and operational maturity.

12 Issuer decision tree

Figure 3 translates the framework into an issuer-usable sequence of questions. The purpose is not to force linearity in all real implementations, but to discipline decision-making so that legal and operational feasibility precede ecosystem optimization.

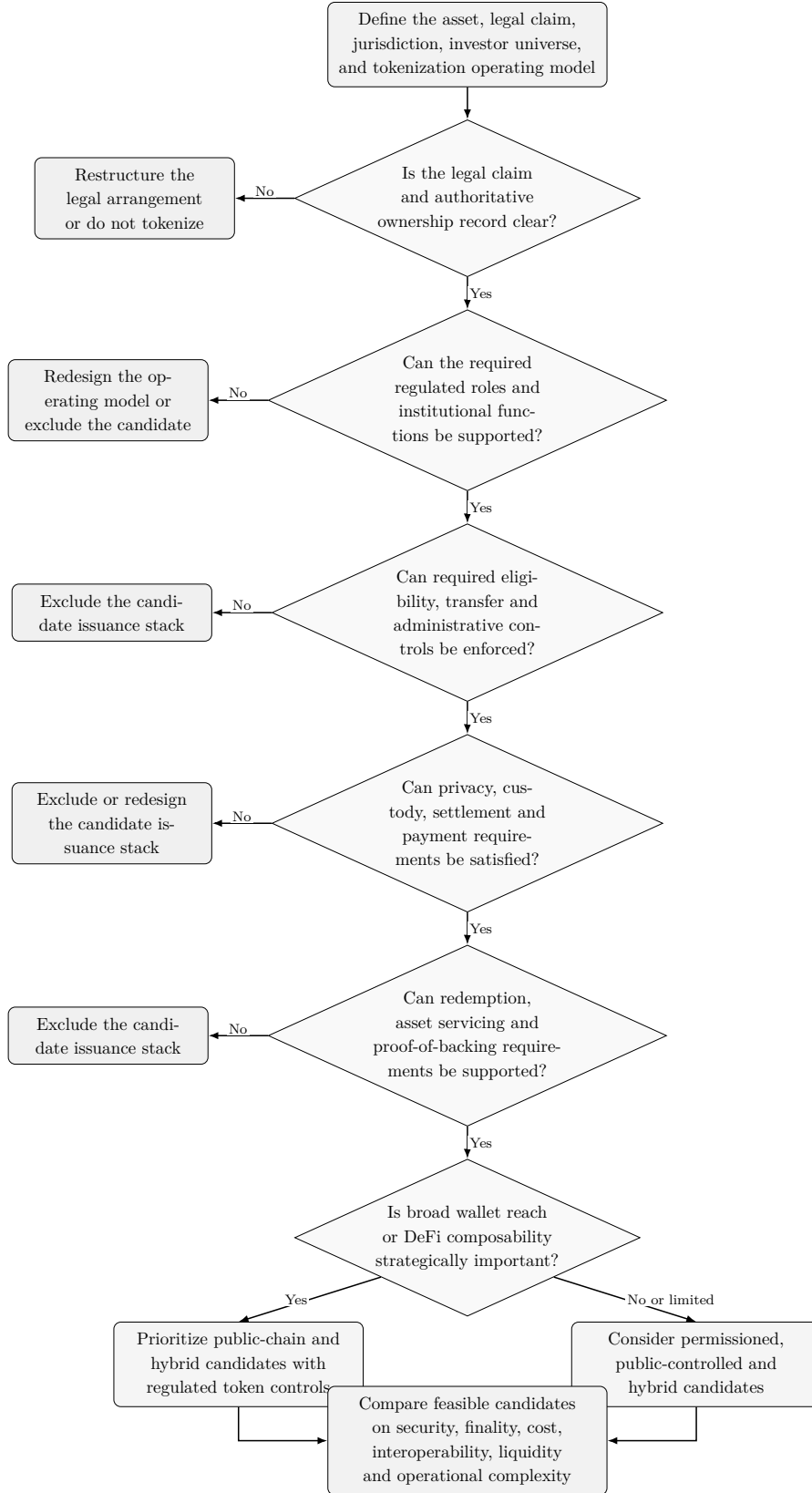


Figure 3: Issuer decision tree for evaluating feasible RWA issuance stacks

Table 12: Issuer due-diligence checklist

Due-diligence question	Status	Required determination
Is the token’s legal classification and holder claim defined?	☐	Instrument type, governing law, issuer or obligor, holder rights, claim priority, and insolvency treatment.
Is the authoritative ownership or entitlement record identified?	☐	Legal status of the on-chain and off-chain records, reconciliation, correction, and conflict-resolution procedures.
Are all required regulatory permissions and institutional roles identified?	☐	Issuer, distributor, custodian, venue, transfer agent, administrator, registrar, trustee, auditor, and other regulated providers.
Are investor eligibility, distribution, holding, and transfer rules defined?	☐	Investor categories, jurisdictions, offering restrictions, sanctions controls, and continuing eligibility requirements.
Are administrative intervention powers legally authorized and governed?	☐	Pause, freeze, correction, recovery, or forced transfer; authorized actors, triggers, approvals, safeguards, and audit trail.
Are custody, key management, and asset segregation arrangements adequate?	☐	Token and underlying-asset custody, signing authority, recovery, liability, segregation, and insolvency protection.
Are issuance, settlement, and payment processes defined?	☐	Subscription, pricing, token allocation, cash-leg asset, delivery-versus- payment, settlement finality, failure handling, and reconciliation.
Are servicing, redemption, and lifecycle events documented?	☐	Distributions, corporate actions, valuation, eligible redeemers, price, fees, minimums, timing, suspension, cancellation, and asset delivery.
For backed products, is reserve assurance sufficient?	☐	Asset existence, title or control, segregation, encumbrances, valuation, reconciliation, supply integrity, and evidence frequency.
Are privacy, disclosure, reporting, and recordkeeping obligations addressed?	☐	Data minimization, selective access, required disclosures, audit records, retention, correction, and supervisory reporting.
Are market access and liquidity claims supported by evidence?	☐	Permitted venues, investor access, turnover, market depth, price discovery, market making, collateral use, and redemption capacity.
Is every required capability assigned to a responsible layer and entity?	☐	Base ledger, token contract, identity service, oracle, payment rail, custodian, administrator, venue, or other institution.
Are resilience, monitoring, upgrades, and exit arrangements documented?	☐	Incident response, continuity, contract upgrades, provider substitution, data portability, migration, wind-down, and allocation of responsibility.

Note: Completion requires documented evidence, responsible ownership, and legal or technical validation where appropriate. A checked item should not indicate that the issue has merely been discussed.

13 Worked application: selecting an issuance stack for tokenized private credit

This section demonstrates how the proposed framework can be applied to a specific tokenization decision. The purpose is not to identify a blockchain that is universally optimal for private credit. Rather, it shows how the selection outcome follows from the legal claim, investor population, confidentiality requirements, servicing model, settlement design, and strategic objectives of the

proposed product.

13.1 Use-case definition

Consider an asset manager that intends to tokenize interests in a bankruptcy-remote special-purpose vehicle (SPV) holding a diversified portfolio of senior secured corporate loans. The token represents a proportionate contractual or beneficial interest in the SPV. Distribution is limited to professional, accredited, or otherwise eligible institutional investors.

The underlying loans remain subject to conventional origination, collateral perfection, credit assessment, servicing, and enforcement processes. A regulated administrator maintains the authoritative investor register, while a loan servicer collects borrower payments, monitors covenants, manages arrears, and allocates cash through the applicable payment waterfall. The blockchain supports the representation and transfer of investor positions, but it does not replace the loan agreements, security interests, servicer records, or the legally authoritative register.

Table 13 defines the assumptions used in the worked application. A change in these assumptions may produce a different selection result.

Table 13: Assumptions for the tokenized private-credit use case

Decision element	Use-case assumption	Framework implication
Legal claim	The token represents an interest in an SPV or private-credit fund holding senior secured corporate loans.	The token must be reconciled with the governing documents and authoritative investor register.
Investor population	Distribution is limited to verified institutional investors.	Holding and transfer require continuing identity, jurisdiction, sanctions, and eligibility controls.
Transfer model	Transfers are permitted only between approved investors and may require administrator approval.	Allowlisting, transfer validation, freezing, recovery, and forced-transfer functions are mandatory.
Confidentiality	Borrower identities, loan terms, covenant performance, valuations, and investor positions are commercially sensitive.	Need-to-know data visibility is treated as a mandatory feasibility gate.
Servicing	An external servicer collects payments, monitors covenants, manages defaults, and calculates allocation waterfalls.	The ledger must integrate with off-chain servicing and administration systems.
Redemption and liquidity	Redemptions occur on specified dealing dates and may be subject to notice periods, gates, or suspension. No continuous secondary liquidity is promised.	Redemption capacity matters more than public-exchange or permissionless-DeFi access.
Settlement	Subscriptions and distributions use regulated bank money, tokenized deposits, or another legally acceptable institutional settlement asset.	The payment rail and legal finality of the cash leg must be assessed separately from ledger confirmation.
Strategic objective	The issuer seeks future interoperability with institutional collateral, payment, and portfolio-management applications while retaining confidentiality.	Cross-application interoperability becomes an important Stage 2 criterion after the privacy gate has been satisfied.

Private credit differs from publicly traded debt because its lifecycle depends heavily on confidential borrower information, bilateral contractual terms, periodic valuation, servicing, covenant administration, and default or workout processes. These requirements make the complete is-

suance stack more important than generic network throughput or public-wallet availability.

13.2 Stage 1: mandatory feasibility screening

Five candidate architectures are considered:

1. a public Ethereum deployment using ERC-3643 and external identity, custody, administration, and servicing systems;
2. a Stellar-issued regulated asset using authorization, trustline, and clawback controls;
3. a Solana Token-2022 deployment using transfer hooks, confidential transfers, pausing, and administrative extensions;
4. a permissioned Corda application network; and
5. a Canton-based application using participant-specific data visibility and controlled institutional workflows.

Legal validity, regulatory permissions, custody, servicing, settlement, redemption, and reconciliation cannot be established by any network alone. For this illustration, those conditions are assumed to be satisfiable through the SPV, administrator, custodian, servicer, payment provider, and governing documents. The differentiating Stage 1 questions are whether the architecture can enforce investor restrictions and authorized intervention while preserving the required confidentiality.

Table 14 uses three classifications. “Pass” means that the architecture can satisfy the requirement when combined with the stated institutional components. “Conditional” means that material custom development or further validation is required. “Fail” means that the candidate does not satisfy a mandatory assumption of the use case as currently designed.

Table 14: Stage 1 feasibility screening for the private-credit use case

Feasibility gate	Ethereum ERC-3643	Stellar	Solana Token-2022	Corda	Canton
Investor eligibility and transfer control	Pass	Pass	Cond.	Pass	Pass
Recovery and authorized intervention	Pass	Pass	Cond.	Pass	Pass
Need-to-know position visibility	Fail	Fail	Cond.	Pass	Pass
Confidential servicing workflows	Fail	Fail	Cond.	Pass	Pass
Administrator/servicer integration	Cond.	Cond.	Cond.	Pass	Pass
Controlled institutional participation	Cond.	Cond.	Cond.	Pass	Pass
Payment and settlement integration	Cond.	Cond.	Cond.	Cond.	Cond.
Stage 1 outcome	Exclude	Exclude	Exclude (current case)	Retain	Retain

Note: Cond. = Conditional. “Exclude (current case)” means excluded under the confidentiality and operating assumptions of this private-credit scenario.

Ethereum and Stellar are excluded under the stated assumptions because public ledger visibility does not satisfy the mandatory confidentiality requirement. Identity and transfer-control functions determine who may hold or receive the asset, but they do not by themselves make investor positions, transaction relationships, or servicing events visible only to authorized parties.

Solana is classified as conditional because Token-2022 can support compliance logic and confidential-transfer functionality. However, the existence of these extensions does not by itself establish end-to-end confidentiality for borrower documentation, covenant events, valuations, servicing records, investor identity, and administrator actions. Extension compatibility, institutional wallet support, custody, auditor access, recovery, and production maturity would need to be validated before the architecture could pass the privacy gate.

Corda and Canton proceed to Stage 2 because both architecture families can restrict information to relevant participants.

13.3 Stage 2: comparative assessment of feasible alternatives

Stage 2 compares Corda and Canton only. The purpose is not to create a universal ranking, but to identify which retained architecture provides the stronger fit with the stated priorities.

Table 15: Stage 2 comparison of feasible private-credit architectures

Criterion	Corda	Canton	Use-case implication
Confidentiality	Strong participant-specific visibility within a permissioned application network.	Strong need-to-know disclosure at the contract and workflow level.	Both satisfy the mandatory confidentiality threshold.
Participant governance	Well suited to a defined application network with controlled membership and contractual onboarding.	Supports controlled participation through application and synchronizer governance.	Both are suitable for institutional investors and regulated service providers.
Private-credit lifecycle integration	Strong fit for bilateral or consortium workflows involving lenders, borrowers, agents, and servicers.	Strong fit for multi-party workflows involving issuers, administrators, custodians, servicers, and settlement applications.	Both can support issuance, servicing, distributions, transfers, and redemptions.
Cross-application interoperability	Most direct where participants use the same application network or establish specific external integrations.	Designed to support workflows across independently operated applications and synchronizers.	Canton has the stronger fit when future connectivity to tokenized cash, collateral, or portfolio applications is a strategic objective.
Public distribution and wallet reach	Limited relative to major public blockchains.	Limited and institutionally controlled relative to major public blockchains.	This is not decisive because the product is restricted to institutional investors.
Implementation complexity	Requires application development, network governance, participant onboarding, and integration with existing systems.	Requires workflow development, participant and synchronizer configuration, and integration with existing systems.	Neither architecture should be selected solely on base-layer transaction fees.
Operational fit	Strongest for a relatively closed consortium with stable participants and purpose-built workflows.	Strongest where confidentiality must coexist with broader institutional interoperability.	The stated strategic objective favors Canton.

13.4 Decision and selected issuance stack

Under the stated assumptions, the preferred architecture is a **Canton-based institutional issuance stack**. The application would initially operate through a privately governed synchronizer or another controlled deployment, while preserving the option to connect selected

workflows to compatible institutional applications. The selection is driven by need-to-know privacy, multi-party workflow support, and the strategic objective of future interoperability. It is not based on a claim that Canton is universally cheaper, faster, or more liquid than every alternative.

Table 16: Selected issuance stack for the tokenized private-credit product

Stack component	Proposed implementation and responsibility
Legal instrument	An SPV or private-credit fund issues the legally enforceable interest under defined governing law, offering documents, claim priority, and insolvency treatment.
Authoritative record	A regulated administrator or registrar maintains the controlling investor register and reconciles it with token positions.
Ledger and workflow layer	A Canton-based application records approved issuance, transfers, distributions, redemptions, and administrative events using need-to-know data visibility.
Investor identity	An authorized onboarding provider verifies investor category, beneficial ownership, jurisdiction, sanctions status, and continuing eligibility. Sensitive identity evidence remains off-ledger where appropriate.
Transfer compliance	Application logic verifies the eligibility of both parties before transfer and requires any necessary administrator or issuer approval.
Administrative controls	Authorized parties can pause transfers, block positions, correct errors, recover access, or execute legally valid forced transfers under documented governance and audit procedures.
Underlying-asset custody and control	The SPV, trustee, security agent, or custodian maintains legally effective control over loan assets, collateral rights, and relevant accounts.
Loan servicing	The appointed servicer collects borrower payments, monitors covenants, administers arrears and defaults, and supplies approved lifecycle data to the administrator and ledger application.
Valuation and reporting	The administrator or valuation agent calculates portfolio value and investor entitlements. Only approved results or proofs are distributed to relevant parties.
Cash settlement	Subscriptions, distributions, and redemptions use regulated bank money, tokenized deposits, or another legally acceptable settlement asset. The payment leg is reconciled with the corresponding token event.
Reserve and position reconciliation	Token supply and investor positions are reconciled with the authoritative register, SPV records, underlying loan balances, cash accounts, and outstanding liabilities.
Redemption and wind-down	The governing documents specify eligible redeemers, dealing dates, notice periods, gates, fees, suspensions, token cancellation, asset realization, data migration, and provider-replacement procedures.

The selected network therefore delivers only part of the required operating model. Legal rights remain dependent on the SPV and governing documents; credit quality remains dependent on origination and underwriting; asset performance remains dependent on borrowers and the servicer; and settlement remains dependent on the payment infrastructure. The network is selected because its technical characteristics fit these institutional arrangements, not because it replaces them.

13.5 Sensitivity of the recommendation

The recommendation changes when the use-case assumptions change. If borrower and servicing data remain entirely off-chain, public visibility of token balances is acceptable, and public-chain collateral connectivity is a strategic priority, an Ethereum deployment using a regulated token standard could pass Stage 1 and become preferable. If the product is operated by a small and stable consortium and does not require broader cross-application interoperability, Corda may provide an equally strong or stronger fit. Solana or Stellar could also become feasible where

confidentiality requirements are narrower and their administrative controls, custody support, and integration arrangements are independently validated.

The defensible conclusion is therefore not that Canton is the best blockchain for all private-credit tokenization. Rather, for an institutional private-credit product requiring need-to-know confidentiality, restricted transfers, administrator and servicer integration, and future interoperability with institutional financial applications, a Canton-based issuance stack provides the strongest contextual fit among the evaluated alternatives.

14 Discussion

The review reframes blockchain selection for tokenized real-world assets as a problem of institutionally complete issuance-stack design. This differs from approaches that compare networks primarily through throughput, fees, transaction latency, or ecosystem size. Such characteristics remain relevant, but only after the legal claim, authoritative ownership record, investor restrictions, custody structure, settlement process, lifecycle administration, redemption mechanism, and evidentiary arrangements have been shown to be feasible.

The findings also extend Task–Technology Fit from an individual information-systems context to an interorganizational financial-infrastructure setting. In this application, the task is not merely the execution of a blockchain transaction. It encompasses issuance, investor onboarding, transfer validation, settlement, servicing, reporting, redemption, recovery, and governance. Correspondingly, the relevant technology is not the base ledger alone but the complete configuration of ledger, token standard, identity services, smart contracts, custody, administration, payment infrastructure, and legal institutions. Fit therefore depends on whether the combined stack can perform the required tasks under the applicable legal and operational constraints.

The feasibility-first structure is an important practical implication of the framework. Conventional weighted comparisons may permit strong performance on cost, throughput, or ecosystem reach to compensate for the absence of a legally required capability. The proposed framework rejects such compensation. Failure to support an authoritative ownership record, legally required investor restrictions, acceptable settlement, or enforceable redemption excludes the candidate before comparative advantages are considered.

The private-credit application illustrates this logic. Public-chain architectures provide meaningful benefits in wallet availability, programmability, and ecosystem connectivity, but those advantages do not compensate for the failure to satisfy the scenario’s mandatory confidentiality assumptions. Corda and Canton proceed because their architectures support participant-specific information sharing and controlled institutional workflows. Canton is preferred only because the stated use case also assigns strategic importance to future cross-application interoperability. A different confidentiality, distribution, settlement, or interoperability profile could favour Corda, Ethereum, Stellar, Solana, or another architecture.

The framework also has implications for regulators and market-infrastructure providers. Regulatory analysis should distinguish between capabilities provided by the ledger and those delivered by token contracts, identity providers, custodians, administrators, payment systems, and governing documents. Similarly, issuers should avoid describing public observability, token

supply, total value locked, or transfer volume as evidence of legal certainty, reserve quality, or secondary-market liquidity. These outcomes must be evaluated independently across the complete issuance stack.

15 Limitations

This article has four limitations. First, the evidence base spans heterogeneous source types, so synthesis necessarily balances peer-reviewed evidence with official technical and institutional documentation. Second, the market remains dynamic; some implementation details may evolve faster than journal publication cycles. Third, the framework is still conceptual which need empirically tested through a scored multicase dataset. Fourth, while the review covers multiple jurisdictions, legal analysis remains high-level and should not substitute for jurisdiction-specific legal advice on issuance structuring, securities law, tax, custody, or insolvency issues.

16 Future research agenda

Future research should move from conceptual fit to observed fit. Several promising directions emerge from the review: longitudinal studies of migration from pilot to production; event studies on mint, redeem, and transfer behavior around market stress; comparative studies of public-chain versus permissioned-stack operating costs; legal-comparative work on authoritative records and insolvency treatment; and market-structure studies separating primary issuance from genuine secondary liquidity (Luo et al. 2026; Vella et al. 2026; Agur and Copestake 2025; Mafrur and Khadijah 2026; Mafrur 2026).

Another important frontier is evidentiary design. Proof-of-backing, NAV reporting, cash-leg quality, and reserve attestations remain under-theorized in the current literature. Research should examine how attestation regimes, oracle structures, and legal recourse interact in different asset classes, and should distinguish strongly between transparency, verification, and enforceability (Chainlink 2025a; Circle 2026a; Paxos 2025; International Organization of Securities Commissions 2025).

17 Conclusion

Selecting a blockchain network for tokenized real-world assets is not a matter of choosing the chain with the biggest ecosystem, the lowest fees, or the highest throughput. It is a problem of institutional and technological fit. The relevant decision unit is the complete issuance stack, and the correct sequence is to screen for feasibility before comparing preferences.

Using a systematic integrative review and Task–Technology Fit as the principal theoretical lens, this article showed that suitability depends on asset type, legal claim, jurisdiction, issuance structure, investor eligibility, transfer restrictions, issuer-control needs, custody, settlement, redemption, privacy, liquidity infrastructure, interoperability, security, and proof of backing. It also showed that many highly salient capabilities are not native blockchain features but are instead delivered through token standards, contracts, middleware, or external institutions. The

resulting framework offers issuers a disciplined way to move from requirements to architecture without assuming that one blockchain is universally superior.

A Appendices

Appendix A: Sample search strings

Table 17: Sample search strings used in source identification

Search family	Sample formulation
RWA concept and review	"tokenized real-world assets" OR "RWA tokenization" review OR systematic
Network selection	"blockchain network selection" tokenized securities OR tokenized funds
Legal structure	tokenized securities ownership record transfer agent custody redemption
Asset-class specific	tokenized treasuries / tokenized money market fund / tokenized gold / private credit tokenization
Standards and controls	ERC-3643 / ERC-1450 / SEP-24 / Token-2022 / clawback / proof of reserve
Settlement and interoper- ability	tokenized assets settlement stablecoin tokenized deposits DvP interoperability

Appendix B: Supplementary source map used during coding

The following source clusters were used during coding and are cited here to make the evidence base fully explicit.

Foundational theory and review-method sources: (Goodhue and Thompson 1995; Zigurs and Buckland 1998; Dishaw and Strong 1999; Donaldson 2001; DiMaggio and Powell 1983; Tornatzky and Fleischer 1990; Page et al. 2021; Whitemore and Knafl 2005; Torraco 2005; Webster and Watson 2002; Braun and Clarke 2006).

Foundational blockchain, fintech, and architecture literature: (Nakamoto 2008; Buterin 2014; Croman et al. 2016; Garay et al. 2015; Lewenberg et al. 2015; Yli-Huumo et al. 2016; Casino et al. 2019; Xu et al. 2017; Catalini and Gans 2020; Y. Chen and Bellavitis 2020; Cong et al. 2021; Schär 2021; Gomber et al. 2018; Lee and Shin 2018; Arner et al. 2015; Puschmann 2017; Belchior et al. 2021; Al-Breiki et al. 2020; Caldarelli 2020; Bünz et al. 2020; Cohny et al. 2019; De Filippi and Wright 2018; Lessig 1999; Amihud and Mendelson 1986; Campbell and Shiller 1988; Soto 2000).

RWA-focused academic and working-paper corpus: (Avci and Erzurumlu 2023; Kona-shevyeh 2020; X. Li et al. 2019; S. Chen et al. 2024; Baltais et al. 2024; Laschinger et al. 2026; Avgouleas and Kiayias 2020; Bullmann et al. 2019; J. Li 2025; Luo et al. 2026; Chemla and Tinn 2026; Vella et al. 2026; Ndukaji 2026; Mafrur 2025; Mafrur and Khadijah 2026; Mafrur 2026; Sinha et al. 2025; Agur and Copestake 2025; Agur, Villegas-Bauer, et al. 2025; Adrian 2026; Adrian et al. 2026).

Regulatory and institutional corpus: (Bank for International Settlements 2023; Aldasoro et al. 2023; Basel Committee on Banking Supervision 2022; Committee on Payments and Market Infrastructures 2024; Bank for International Settlements 2025b; Bank for International Settlements 2025a; Bank for International Settlements 2026; Financial Stability Board 2023; Financial Stability Board 2024; International Organization of Securities Commissions 2020; International Organization of Securities Commissions 2023; International Organization of Securities Commissions 2025; Kaousar Nassr 2021; Organisation for Economic Co-operation and Development 2025; Organisation for Economic Co-operation and Development 2021; World Bank 2020; World Bank 2023a; World Bank 2023b; European Central Bank 2026a; European Central Bank 2025b; European Central Bank 2025a; European Central Bank 2026b; Monetary Authority of Singapore 2022; Monetary Authority of Singapore 2024b; Monetary Authority of Singapore 2024a; Monetary Authority of Singapore 2025c; Monetary Authority of Singapore 2025b; Monetary Authority of Singapore 2025a; Monetary Authority of Singapore 2025d; U.S. Securities and Exchange Commission 2026; U.S. Securities and Exchange Commission 2025c; Investor.gov 2025; European Union 2022; European Union 2023).

Issuer, product, settlement, and methodology documentation: (U.S. Securities and Exchange Commission 2021; U.S. Securities and Exchange Commission 2025b; Franklin Templeton 2026; U.S. Securities and Exchange Commission 2024; U.S. Securities and Exchange Commission 2025a; Securitize 2024b; Securitize 2024a; Ondo Finance 2026a; Ondo Finance 2026b; Circle 2026d; Circle 2026c; Circle 2026a; Circle 2026b; Paxos 2019; Paxos 2025; Paxos 2026; Chainlink 2025b; Chainlink 2025a; SWIFT 2025; SWIFT 2026b; SWIFT 2026a; RWA.xyz 2026c; RWA.xyz 2026a; RWA.xyz 2026b; DefiLlama 2026).

Technical and infrastructure documentation used to code issuance-stack capabilities: (Tokeny 2023; Tokeny 2026; Vogelsteller and Buterin 2015; Dossa et al. 2018; Marks et al. 2018; Kupriianov and Svirsky 2018; Lebrun et al. 2021; Lo Buglio et al. 2025; Stellar Development Foundation 2026; Stellar Development Foundation 2025b; Stellar Development Foundation 2025a; Stellar Development Foundation 2025d; Stellar Development Foundation 2025c; Solana Foundation 2026a; Solana Foundation 2026b; Solana Foundation 2024; Ava Labs 2023; Ava Labs 2024; Ava Labs 2026; Hearn and Brown 2019; R3 2024a; R3 2023; R3 2024b; Digital Asset 2026a; Digital Asset 2026b; Digital Asset 2026c; Hyperledger 2019; Hyperledger Besu 2026; Fireblocks 2026; Fireblocks 2025; BitGo 2026; BitGo 2025; Apex Group 2026; Securitize 2026; Aave Companies 2022; Adams et al. 2021; Centrifuge 2021; Centrifuge 2025; Lambur et al. 2018; Benligiray et al. 2020; Breidenbach et al. 2021).

References

- Aave Companies (2022). *Aave Protocol V3 Technical Paper*. URL: <https://github.com/aave/protocol-v3>.
- Adams, Hayden, Noah Zinsmeister, Moody Salem, River Keefer, and Dan Robinson (2021). *Uniswap V3 Core*. URL: <https://uniswap.org/whitepaper-v3.pdf>.
- Adrian, Tobias (2026). *Tokenized Finance*. IMF Notes No. 2026/001. International Monetary Fund.
- Adrian, Tobias, Tommaso Mancini-Griffoli, and Yaiza Cabedo (2026). *The Rise of Tokenization: Deciphering New Trends in Payments and Asset Tokenization*. IMF Notes No. 2026/006.

- International Monetary Fund. URL: <https://www.imf.org/en/publications/imf-notes/issues/2026/07/01/the-rise-of-tokenization-deciphering-new-trends-in-payments-and-asset-tokenization-577138>.
- Agur, Itai and Alexander Copestake (2025). *Optimal Policy for Financial Market Tokenization*. Working Paper No. 2025/185. International Monetary Fund.
- Agur, Itai, Germán Villegas-Bauer, Tommaso Mancini-Griffoli, Maria Soledad Martinez Peria, and Brandon Tan (2025). *Tokenization and Financial Market Inefficiencies*. Fintech Notes No. 2025/001. International Monetary Fund.
- Aldasoro, Iñaki, Sebastian Doerr, Leonardo Gambacorta, Rodney Garratt, and Priscilla Koo Wilkens (2023). *The Tokenisation Continuum*. Bank for International Settlements.
- Amihud, Yakov and Haim Mendelson (1986). “Asset Pricing and the Bid-Ask Spread”. In: *Journal of Financial Economics*.
- Apex Group (2026). *Digital Asset Fund Services & Administration*. URL: <https://www.apexgroup.com/fund-solution/digital-asset-fund-services/>.
- Arner, Douglas W., Janos Barberis, and Ross P. Buckley (2015). “The Evolution of Fintech: A New Post-Crisis Paradigm”. In: *Georgetown Journal of International Law*.
- Ava Labs (2023). *Avalanche Launches Evergreen for Institutional Blockchain Deployments*. URL: <https://www.avax.network/about/blog/avalanche-launches-evergreen-for-institutional-blockchain-deployments/>.
- (2024). *Citi Tests Benefits of Private Markets Tokenization with Avalanche Evergreen Subnet Spruce*. URL: <https://www.avax.network/about/blog/citi-tests-benefits-of-private-markets-tokenization-with-avalanche-evergreen-subnet-spruce>.
- (2026). *Institutions and Capital Markets*. URL: <https://www.avax.network/institutions>.
- Avci, Gulsah and Yonca O. Erzurumlu (2023). “Blockchain Tokenization of Real Estate Investment: A Security Token Offering Procedure and Legal Design Proposal”. In: *Journal of Property Research*.
- Avgouleas, Emiliós and Aggelos Kiayias (2020). *The Architecture of Decentralised Finance Platforms: A New Open Finance Paradigm*. SSRN.
- Baltais, Monika, Elina Sondore, Talis J. Putniņš, and Jens R. Karlsen (2024). *Economic Impact Potential of Real-World Asset Tokenization*. UTS Business School, University of Technology Sydney.
- Bank for International Settlements (2023). *Blueprint for the Future Monetary System: Improving the Old, Enabling the New*. Bank for International Settlements. URL: <https://www.bis.org/publ/arpdf/ar2023e3.pdf>.
- (2025a). *Annual Economic Report 2025, Chapter III: The Next-Generation Monetary and Financial System*. Bank for International Settlements. URL: <https://www.bis.org/publ/arpdf/ar2025e3.pdf>.
- (2025b). *Leveraging Tokenisation for Payments and Financial Transactions*. Bank for International Settlements. URL: <https://www.bis.org/publ/othp92.pdf>.
- (2026). *Annual Economic Report 2026, Chapter III: Anchoring Trust in Money: Innovation Beyond Stablecoins*. Bank for International Settlements. URL: <https://www.bis.org/publ/arpdf/ar2026e3.htm>.
- Basel Committee on Banking Supervision (2022). *Prudential Treatment of Cryptoasset Exposures*. Bank for International Settlements.

- Belchior, Rafael, André Vasconcelos, Sérgio Guerreiro, and Miguel Correia (2021). “A Survey on Blockchain Interoperability: Past, Present, and Future Trends”. In: *ACM Computing Surveys*.
- Benligiray, Burak, Sasa Milic, and Heikki Vanttinen (2020). *API3: Decentralized APIs for Web 3.0*. URL: <https://github.com/api3dao/whitepaper>.
- BitGo (2025). *Digital Asset Custody*. URL: <https://www.bitgo.com/uk/resources/blog/digital-asset-custody/>.
- (2026). *Crypto Custody for Institutions: Qualified Custody*. URL: <https://www.bitgo.com/products/qualified-custody/>.
- Braun, Virginia and Victoria Clarke (2006). “Using Thematic Analysis in Psychology”. In: *Qualitative Research in Psychology*.
- Breidenbach, Lorenz, Christian Cachin, Bennett Chan, Alex Coventry, Steve Ellis, Ari Juels, Farinaz Koushanfar, Andrew Miller, Brendan Magauran, Dmitri Moroz, et al. (2021). *Chainlink 2.0: Next Steps in the Evolution of Decentralized Oracle Networks*. Chainlink Labs.
- Al-Breiki, Hamda, Muhammad Habib Ur Rehman, Khaled Salah, and Davor Svetinovic (2020). “Trustworthy Blockchain Oracles: Review, Comparison, and Open Research Challenges”. In: *IEEE Access*.
- Bullmann, Dirk, Jonas Klemm, and Andrea Pinna (2019). *In Search of Stability in Crypto-Assets: Are Stablecoins the Solution?* Occasional Paper. European Central Bank.
- Bünz, Benedikt, Shashank Agrawal, Mahdi Zamani, and Dan Boneh (2020). “Zether: Towards Privacy in a Smart Contract World”. In: *Financial Cryptography and Data Security*.
- Buterin, Vitalik (2014). *A Next-Generation Smart Contract and Decentralized Application Platform*. URL: <https://ethereum.org/whitepaper/>.
- Caldarelli, Guido (2020). “Understanding the Blockchain Oracle Problem: A Call for Action”. In: *Information*.
- Campbell, John Y. and Robert J. Shiller (1988). “The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors”. In: *The Review of Financial Studies*.
- Casino, Fran, Thomas K. Dasaklis, and Constantinos Patsakis (2019). “A Systematic Literature Review of Blockchain-Based Applications: Current Status, Classification and Open Issues”. In: *Telematics and Informatics*.
- Catalini, Christian and Joshua S. Gans (2020). “Some Simple Economics of the Blockchain”. In: *Communications of the ACM*.
- Centrifuge (2021). *Centrifuge Protocol Whitepaper*. URL: <https://www.securities.io/centrifuge-whitepaper/>.
- (2025). *Centrifuge: Real-World Asset Financing*. URL: <https://centrifuge.io/>.
- Chainlink (2025a). *7 Key Principles for Proof of Reserves*. URL: <https://chain.link/blog/7-key-principles-for-proof-of-reserves>.
- (2025b). *Proof of Reserve*. URL: <https://chain.link/proof-of-reserve>.
- Chemla, Gilles and Katrin Tinn (2026). *Digital Ownership: The Tokenization of Real-World Assets*. SSRN.
- Chen, Siyu, Meng Jiang, and Xiaowei Luo (2024). “Exploring the Security Issues of Real World Assets (RWA)”. In: *Proceedings of the Workshop on Decentralized Finance and Security*.
- Chen, Yan and Cristiano Bellavitis (2020). “Blockchain Disruption and Decentralized Finance: The Rise of Decentralized Business Models”. In: *Journal of Business Venturing Insights*.
- Circle (2026a). *Transparency & Stability*. URL: <https://www.circle.com/transparency>.

- Circle (2026b). *USDC*. URL: <https://www.circle.com/usdc>.
- (2026c). *USYC Documentation*. URL: <https://usyc.docs.hashnote.com/>.
- (2026d). *USYC: Tokenized Money Market Fund*. URL: <https://www.circle.com/usyc>.
- Cohney, Shaanan, David Hoffman, Jeremy Sklaroff, and David Wishnick (2019). “Coin-Operated Capitalism”. In: *Columbia Law Review*.
- Committee on Payments and Market Infrastructures (2024). *Tokenisation in the Context of Money and Other Assets*. Bank for International Settlements. URL: <https://www.bis.org/cpmi/publ/d225.pdf>.
- Cong, Lin William, Ye Li, and Neng Wang (2021). “Tokenomics: Dynamic Adoption and Valuation”. In: *The Review of Financial Studies*.
- Croman, Kyle, Christian Decker, Ittay Eyal, Adem Efe Gencer, Ari Juels, Ahmed Kosba, Andrew Miller, Prateek Saxena, Elaine Shi, Emin Gün Sirer, et al. (2016). “On Scaling Decentralized Blockchains: A Position Paper”. In: *International Conference on Financial Cryptography and Data Security*.
- De Filippi, Primavera and Aaron Wright (2018). *Blockchain and the Law: The Rule of Code*. Harvard University Press.
- DefiLlama (2026). *DefiLlama and Our Methodology*. URL: <https://docs.llama.fi/>.
- Digital Asset (2026a). *Canton Network Overview*. URL: <https://docs.digitalasset.com/integrate/devnet/canton-network-overview/index.html>.
- (2026b). *Canton Network Quickstart Observability & Troubleshooting Overview*. URL: <https://docs.digitalasset.com/build/3.4/quickstart/observe/observability-troubleshooting-overview.html>.
- (2026c). *xReserve Workflows*. URL: <https://docs.digitalasset.com/usdc/xreserve/workflows.html>.
- DiMaggio, Paul J. and Walter W. Powell (1983). “The Iron Cage Revisited: Institutional Isomorphism and Collective Rationality in Organizational Fields”. In: *American Sociological Review*.
- Dishaw, Mark T. and Diane M. Strong (1999). “Extending the Technology Acceptance Model with Task-Technology Fit Constructs”. In: *Information & Management*.
- Donaldson, Lex (2001). *The Contingency Theory of Organizations*. Sage.
- Dossa, Adam, G., and Fabian Vogelsteller (2018). *EIP-1400: Security Token Standard*. URL: <https://eips.ethereum.org/>.
- European Central Bank (2025a). *Pontes*. URL: <https://www.ecb.europa.eu/paym/target/pontes/html/index.en.html>.
- (2025b). *The Eurosystem’s Exploratory Work on New Technologies for Wholesale Central Bank Money Settlement*. European Central Bank. URL: <https://www.ecb.europa.eu/pub/pdf/other/ecb.exploratoryworknewtechnologies202506.en.pdf>.
- (2026a). *Tokenisation and DLT*. URL: <https://www.ecb.europa.eu/paym/dlt/html/index.en.html>.
- (2026b). “Towards an Efficient and Integrated Digital Capital Market in Europe: The Role of Tokenisation and the Eurosystem’s Policy Response”. In: *Macroprudential Bulletin*. URL: https://www.ecb.europa.eu/press/financial-stability-publications/macroprudential-bulletin/html/ecb.mpbu202604_02.en.html.

- European Union (2022). *Regulation (EU) 2022/858 on a Pilot Regime for Market Infrastructures Based on Distributed Ledger Technology*. URL: <https://eur-lex.europa.eu/eli/reg/2022/858/oj/eng>.
- (2023). *Regulation (EU) 2023/1114 on Markets in Crypto-Assets*. URL: <https://eur-lex.europa.eu/eli/reg/2023/1114/oj/eng>.
- Financial Stability Board (2023). *The Financial Stability Risks of Decentralised Finance*. Financial Stability Board. URL: <https://www.fsb.org/uploads/P160223.pdf>.
- (2024). *The Financial Stability Implications of Tokenisation*. Financial Stability Board. URL: <https://www.fsb.org/uploads/P221024-2.pdf>.
- Fireblocks (2025). *Unlocking the Next Wave of Tokenized Assets*. Fireblocks. URL: <https://www.fireblocks.com/report/unlocking-the-next-wave-of-tokenized-assets>.
- (2026). *Tokenization*. URL: <https://www.fireblocks.com/products/tokenization>.
- Franklin Templeton (2026). *Franklin OnChain U.S. Government Money Fund (FOBXX)*. URL: <https://www.franklintempleton.com/investments/options/money-market-funds/products/29386/SINGLCLASS/franklin-on-chain-u-s-government-money-fund/FOBXX>.
- Garay, Juan A., Aggelos Kiayias, and Nikos Leonardos (2015). “The Bitcoin Backbone Protocol: Analysis and Applications”. In: *Annual International Conference on the Theory and Applications of Cryptographic Techniques*.
- Gomber, Peter, Jasmin-Amy Koch, and Michael Siering (2018). “On the FinTech Revolution: Interpreting the Forces of Innovation, Disruption, and Transformation in Financial Services”. In: *Journal of Management Information Systems*.
- Goodhue, Dale L. and Ronald L. Thompson (1995). “Task-Technology Fit and Individual Performance”. In: *MIS Quarterly*.
- Hearn, Mike and Richard Gendal Brown (2019). *Corda: A Distributed Ledger*. R3. URL: <https://docs.r3.com/en/pdf/corda-technical-whitepaper.pdf>.
- Hyperledger (2019). *Announcing Hyperledger Besu*. URL: <https://www.hyperledger.org/blog/2019/08/29/announcing-hyperledger-besu>.
- Hyperledger Besu (2026). *Besu Wiki: Privacy and Permissioning*. URL: <https://wiki.hyperledger.org/display/BESU>.
- International Organization of Securities Commissions (2020). *Issues, Risks and Regulatory Considerations Relating to Crypto-Asset Trading Platforms*. IOSCO. URL: <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD649.pdf>.
- (2023). *Policy Recommendations for Crypto and Digital Asset Markets*. IOSCO. URL: <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD747.pdf>.
- (2025). *Tokenization of Financial Assets*. IOSCO. URL: <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD809.pdf>.
- Investor.gov (2025). *Transfer Agents*. URL: <https://www.investor.gov/introduction-investing/investing-basics/glossary/transfer-agents>.
- Kaousar Nassr, Iota (2021). *Understanding the Tokenisation of Assets in Financial Markets*. OECD. URL: https://www.oecd.org/content/dam/oecd/en/publications/reports/2021/11/understanding-the-tokenisation-of-assets-in-financial-markets_2e657111/c033401a-en.pdf.
- Konashevych, Oleksii (2020). “Constraints and Benefits of the Blockchain Use for Real Estate and Property Rights”. In: *Journal of Property, Planning and Environmental Law*.
- Kupriianov, Maxim and Julian Svirsky (2018). *ERC-1462: Base Security Token*. URL: <https://eips.ethereum.org/EIPS/eip-1462>.

- Lambur, Hart, Allison Lu, and Matt Rice (2018). *UMA: The Optimistic Oracle and Dispute Arbitration*. URL: <https://umaproject.org/>.
- Laschinger, Ralf, Heiko Leonhard, Gregor Dorfleitner, and Wolfgang Schäfers (2026). *Liquidity Mechanisms in Decentralized Finance: Design and Fragmentation in Illiquid Real-World Asset Markets*. SSRN.
- Lebrun, Joachim, Tony Malghem, Kevin Thizy, Luc Falempin, and Adam Boudjemaa (2021). *ERC-3643: T-REX – Token for Regulated Exchanges*. URL: <https://eips.ethereum.org/EIPS/eip-3643>.
- Lee, In and Yong Jae Shin (2018). “Fintech: Ecosystem, Business Models, Investment Decisions, and Challenges”. In: *Business Horizons*.
- Lessig, Lawrence (1999). *Code and Other Laws of Cyberspace*. Basic Books.
- Lewenberg, Yoad, Yonatan Sompolsky, and Aviv Zohar (2015). “Inclusive Block Chain Protocols”. In: *International Conference on Financial Cryptography and Data Security*.
- Li, Jiasun (2025). *Current Landscape of the Real-World Asset (RWA) Tokenization Ecosystem*. SSRN.
- Li, Xiaofeng, Xiaoqing Wu, Xue Pei, and Zhiyu Yao (2019). “Tokenization: Open Asset Protocol on Blockchain”. In: *IEEE International Conference on Information and Computer Technologies*.
- Lo Buglio, Dario, Tino Martinez Molina, and Mihai Colceriu (2025). *ERC-7943: uRWA – Universal Real World Asset Interface*. URL: <https://eips.ethereum.org/EIPS/eip-7943>.
- Luo, Junliang, Xihan Xiong, Zonglun Li, Hong Kang, Xue Liu, William J. Knottenbelt, and Katrina Tinn (2026). *SoK of RWA Tokenization: A Systematization of Concepts, Architectures, and Legal Interoperability*. arXiv preprint arXiv:2604.06608.
- Mafrur, Rischana (2025). *Tokenize Everything, But Can You Sell It? RWA Liquidity Challenges and the Road Ahead*. arXiv preprint arXiv:2508.11651.
- (2026). “Tokenized but Illiquid? Evidence from Real-World Asset Markets”. In: *FinTech 5.3*, p. 62.
- Mafrur, Rischana and Khadijah (2026). *Beyond TVL: An Explainable Risk Scoring Framework for Tokenized Real-World Assets*. arXiv preprint arXiv:2605.29689.
- Marks, Howard, Devender Gollapally, Joe Mathews, Jordan Jahja, John Shiple, and David Zhang (2018). *ERC-1450: RTA-Controlled Security Token*. URL: <https://eips.ethereum.org/EIPS/eip-1450>.
- Monetary Authority of Singapore (2022). *MAS Partners the Industry to Pilot Use Cases in Digital Assets under Project Guardian*. URL: <https://www.mas.gov.sg/news/media-releases/2022/mas-partners-the-industry-to-pilot-use-cases-in-digital-assets>.
- (2024a). *MAS Announces Plans to Support Commercialisation of Asset Tokenisation*. URL: <https://www.mas.gov.sg/news/media-releases/2024/mas-announces-plans-to-support-commercialisation-of-asset-tokenisation>.
- (2024b). *MAS Expands Industry Collaboration to Scale Asset Tokenisation for Financial Services*. URL: <https://www.mas.gov.sg/news/media-releases/2024/mas-expands-industry-collaboration-to-scale-asset-tokenisation-for-financial-services>.
- (2025a). *Guardian Fixed Income Framework*. Monetary Authority of Singapore. URL: <https://www.mas.gov.sg/-/media/mas-media-library/development/fintech/guardian/guardian-fixed-income-framework.pdf>.

- Monetary Authority of Singapore (2025b). *Operationalising Tokenised Funds*. Monetary Authority of Singapore. URL: <https://www.mas.gov.sg/-/media/mas-media-library/development/fintech/guardian/project-guardian-operationalising-tokenised-funds.pdf>.
- (2025c). *Project Guardian: Open and Interoperable Digital Asset Networks*. Monetary Authority of Singapore. URL: <https://www.mas.gov.sg/-/media/mas-media-library/development/fintech/project-guardian/project-guardian-open-interoperable-network.pdf>.
- (2025d). *Use of Tokenised Bank Liabilities for Transaction Banking*. Monetary Authority of Singapore. URL: <https://www.mas.gov.sg/-/media/mas-media-library/development/fintech/guardian/project-guardian-fx-workstream-transaction-banking.pdf>.
- Nakamoto, Satoshi (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. URL: <https://bitcoin.org/bitcoin.pdf>.
- Ndukaji, Adaobi (2026). *Tokenization of Real-World Assets (RWA) and the Democratization of Entrepreneurial Finance*. SSRN.
- Ondo Finance (2026a). *OUSG*. URL: <https://ondo.finance/ousg>.
- (2026b). *OUSG Overview*. URL: <https://docs.ondo.finance/qualified-access-products/ousg/overview>.
- Organisation for Economic Co-operation and Development (2021). *Regulatory Approaches to the Tokenisation of Assets*. OECD. URL: https://www.oecd.org/en/publications/regulatory-approaches-to-the-tokenisation-of-assets_aea35466-en.html.
- (2025). *Tokenisation of Assets and Distributed Ledger Technologies in Financial Markets*. OECD. URL: https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/01/tokenisation-of-assets-and-distributed-ledger-technologies-in-financial-markets_be149012/40e7f217-en.pdf.
- Page, Matthew J., Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, Roger Chou, Julie Glanville, Jeremy M. Grimshaw, Asbjørn Hróbjartsson, Manoj M. Lalu, Tianjing Li, Elizabeth W. Loder, Evan Mayo-Wilson, Steve McDonald, Luke A. McGuinness, Lesley A. Stewart, James Thomas, Andrea C. Tricco, Vivian A. Welch, Penny Whiting, and David Moher (2021). “The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews”. In: *BMJ*.
- Paxos (2019). *Paxos Launches PAX Gold: Physical Gold on the Blockchain*. URL: <https://www.paxos.com/newsroom/paxos-launches-pax-gold-physical-gold-on-the-blockchain>.
- (2025). *PAX Gold Terms and Conditions*. URL: <https://www.paxos.com/terms-and-conditions/pax-gold-terms-conditions>.
- (2026). *PAX Gold*. URL: <https://www.paxos.com/pax-gold>.
- Puschmann, Thomas (2017). “Fintech”. In: *Business & Information Systems Engineering*.
- R3 (2023). *Transaction Privacy Enhancements*. URL: <https://docs.r3.com/en/platform/corda/5.2/developing-applications/ledger/enhanced-ledger-privacy.html>.
- (2024a). *Corda Enterprise*. URL: <https://docs.r3.com/en/platform/corda/4.12/enterprise.html>.
- (2024b). *Two Phase Finality*. URL: <https://docs.r3.com/en/platform/corda/4.11/enterprise/two-phase-finality.html>.
- RWA.xyz (2026a). *Coverage & Eligibility*. URL: <https://docs.rwa.xyz/methodology/coverage>.
- (2026b). *Introducing Our New Asset Classification Framework*. URL: <https://app.rwa.xyz/blog/introducing-our-new-asset-classification-framework>.

- RWA.xyz (2026c). *Market Cap and Total Value Methodology*. URL: <https://docs.rwa.xyz/methodology/market-cap-and-total-value>.
- Schär, Fabian (2021). “Decentralized Finance: On Blockchain- and Smart Contract-Based Financial Markets”. In: *Federal Reserve Bank of St. Louis Review*.
- Securitize (2024a). *BlackRock Launches Its First Tokenized Fund, BUIDL, on the Ethereum Network*. URL: <https://securitize.io/learn/press/blackrock-launches-first-tokenized-fund-buidl-on-the-ethereum-network>.
- (2024b). *Introducing the BlackRock BUIDL Fund*. URL: <https://securitize.io/blackrock/buidl>.
- (2026). *The Leading Tokenization Platform*. URL: <https://securitize.io/>.
- Sinha, Aaryan, Raja Muthalagu, Pranav Pawar, Alavikunhu Panthakkan, and Shadi Atalla (2025). *Blockchain-Powered Asset Tokenization Platform*. arXiv preprint arXiv:2502.06752.
- Solana Foundation (2024). *Token Extensions Enable Native Support for Enterprise-Grade Tokens on Solana*. URL: <https://solana.com/news/token-extensions-on-solana>.
- (2026a). *Token Extensions*. URL: <https://solana.com/docs/tokens/extensions>.
- (2026b). *Token-2022 Program*. URL: <https://spl.solana.com/token-2022>.
- Soto, Hernando de (2000). *The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else*. Basic Books.
- Stellar Development Foundation (2025a). *Anchor Platform*. URL: <https://developers.stellar.org/docs/platforms/anchor-platform>.
- (2025b). *Clawbacks*. URL: <https://developers.stellar.org/docs/build/guides/transactions/clawbacks>.
- (2025c). *Hosted Deposits and Withdrawals (SEP-24)*. URL: <https://developers.stellar.org/docs/platforms/anchor-platform/sep-guide/sep24>.
- (2025d). *Stellar Ecosystem Proposals (SEPs)*. URL: <https://developers.stellar.org/docs/learn/fundamentals/stellar-ecosystem-proposals>.
- (2026). *Assets Overview & Comparison*. URL: <https://developers.stellar.org/docs/tokens/anatomy-of-an-asset>.
- SWIFT (2025). *Swift to Add Blockchain-Based Ledger to Its Infrastructure Stack*. URL: <https://www.swift.com/news-events/press-releases/swift-add-blockchain-based-ledger-its-infrastructure-stack-groundbreaking-move-accelerate-and-scale-benefits-digital-finance>.
- (2026a). *Building the Digital Payment Stack of the Future*. URL: <https://www.swift.com/payments/payment-innovation/blockchain-based-ledger>.
- (2026b). *Swift’s Blockchain-Based Shared Ledger Progresses to MVP Implementation*. URL: <https://www.swift.com/news-events/news/swifts-blockchain-based-shared-ledger-progresses-mvp-implementation>.
- Tokeny (2023). *ERC-3643 Whitepaper*. Tokeny. URL: <https://tokeny.com/wp-content/uploads/2023/05/ERC3643-Whitepaper-T-REX-v4.pdf>.
- (2026). *ERC-3643 Permissioned Tokens Documentation*. URL: <https://docs.tokeny.com/docs/the-erc-3643-token-standard>.
- Tornatzky, Louis G. and Mitchell Fleischer (1990). *The Processes of Technological Innovation*. Lexington Books.
- Torraco, Richard J. (2005). “Writing Integrative Literature Reviews: Guidelines and Examples”. In: *Human Resource Development Review*.

- U.S. Securities and Exchange Commission (2021). *Franklin OnChain U.S. Government Money Fund Filing*. URL: <https://www.sec.gov/Archives/edgar/data/1786958/000137949121001250/filing222856581.htm>.
- (2024). *Form D Filing for BlackRock USD Institutional Digital Liquidity Fund*. URL: <https://www.sec.gov/Archives/edgar/data/2013810/0002014390-24-000001-index.htm>.
- (2025a). *BlackRock Liquidity Funds: DLT Shares Prospectus*. URL: <https://www.sec.gov/Archives/edgar/data/97098/000119312525100851/d100482d485apos.htm>.
- (2025b). *Franklin OnChain U.S. Government Money Fund Prospectus*. URL: <https://www.sec.gov/Archives/edgar/data/1786958/000174177325000031/c485bpos.htm>.
- (2025c). *Transfer Agents*. URL: <https://www.sec.gov/data-research/statistics-data-visualizations/transfer-agents>.
- (2026). *Statement on Tokenized Securities*. URL: <https://www.sec.gov/newsroom/speeches-statements/corp-fin-statement-tokenized-securities-012826-statement-tokenized-securities>.
- Vella, Giorgio, Luca Pennella, and Mark C. Ballandies (2026). *A Taxonomy of Real-World Asset Tokenization for Blockchain-Based Financial Infrastructure*. SSRN.
- Vogelsteller, Fabian and Vitalik Buterin (2015). *ERC-20: Token Standard*. URL: <https://eips.ethereum.org/EIPS/eip-20>.
- Webster, Jane and Richard T. Watson (2002). “Analyzing the Past to Prepare for the Future: Writing a Literature Review”. In: *MIS Quarterly*.
- Whittemore, Robin and Kathleen Knafl (2005). “The Integrative Review: Updated Methodology”. In: *Journal of Advanced Nursing*.
- World Bank (2020). *Blockchain and Associated Legal Issues for Emerging Markets*. World Bank. URL: <https://openknowledge.worldbank.org/entities/publication/bf9189aa-c644-5a99-bf05-e5660a49facd>.
- (2023a). *Infrastructure Tokenization: Does Blockchain Have a Role in the Financing of Infrastructure?* World Bank. URL: <https://openknowledge.worldbank.org/entities/publication/6dbeda37-8d01-4f5c-86a8-f6e6f9cb3d95>.
- (2023b). *Security Token Offerings*. World Bank. URL: <https://documents1.worldbank.org/curated/en/099527312172426888/pdf/IDU1097384031dbe3146e31907719ee423c9318a.pdf>.
- Xu, Xiwei, Ingo Weber, Liming Zhu, Mark Staples, Jan Bosch, Len Bass, Cesare Pautasso, and Paul Rimba (2017). “A Taxonomy of Blockchain-Based Systems for Architecture Design”. In: *2017 IEEE International Conference on Software Architecture (ICSA)*. IEEE, pp. 243–252.
- Yli-Huumo, Jesse, Deokyo Ko, Sujin Choi, Sooyong Park, and Kari Smolander (2016). “Where Is Current Research on Blockchain Technology? A Systematic Review”. In: *PLOS ONE*.
- Zigurs, Ilze and Bonnie K. Buckland (1998). “A Theory of Task/Technology Fit and Group Support Systems Effectiveness”. In: *MIS Quarterly*.